Calculus

MATH 201 Fall 2026

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Grand Valley State University

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Chapter 1

Course Syllabus

1.1 TL;DR

- ***This course is not graded the way you are used to!*** Read Assessment, p. 7, particularly Introduction to Ungrading, p. 7 and Culminating Experience, p. 10. Also check the Important Dates, p. 5!
- Make sure to read Course Policies and Expectations, p. 10, in particular pay attention to Getting More from Class, p. 12.
- Review Classroom and Campus Safety, p. 2 and Safety Information Locations, p. 2.

This document is intended to give you the basics of what you need to know to succeed in this course, including not only details of Content, p. 3, and Course Policies and Expectations, p. 10, but also the guiding principles behind how I approach teaching, and teaching this course in particular.

I reserve the right to adjust this document as needed and will notify you of any changes through Blackboard.

1.2 Instructor and Course Information

1.2.1 Prerequisites

Both MTH 122 and MTH 123; or MTH 124; or assignment through Grand Valley math placement.

1.2.2 Meeting Times

Tuesday and Thursday 8:00–9:50, MAK BL-116.

1.2.3 Contact Information

Table 1.2.1 Instructor and Department Information

Item	Value
Instructor	Dr. Clark Wells
Office Location	MAK Room A2-110 MAK
Office Phone Number	(616) 331-2444
Email	`wellsc@gvsu.edu`
Department Office Location	Room A2-178 MAK
Department Phone Number	(616) 331-2040
Math Tutoring Center	Room A2-601 MAK

Table 1.2.2 Student Hours

Day	Time	Location
Tuesday	2:00–2:50	MAK A2-110
Thursday	2:00–2:50	MAK A2-601 (Math Tutoring Center)

Student Hours are subject to change upon discussion with class members.

1.2.4 Withdrawal Date, Safety, and Accommodation

1.2.4.1 Withdrawal Date

The deadline for withdrawing with a 'W' grade is November 7 at 5:00 PM and is the responsibility of the student.

1.2.4.2 Accessibility, and Accommodation

If you need accommodations due to a learning, physical, or other disability you must present a memo to me from Student Accessibility Resources (SAR) (`gvsu.edu/accessibility`) indicating the existence of a disability and the suggested reasonable accommodations. If you have not already done so, please contact the Student Accessibility Resource (215 The Blue Connection), by calling 331-2490 or emailing `access@gvsu.edu`. Please note that I am unable to provide accommodations based upon disability until I have received a copy of the SAR issued memo. For more information visit `gvsu.edu/accessibility`.

Growth as a person does is certainly not limited to the classroom or academics. If there is an upcoming faith or religious observance that could have an impact on your attendance or work in the class, please submit a request (`gvsu.edu/affirmative/religious-request.htm`) during the first two weeks of the semester, or as soon as is reasonably possible, so that we can plan together accordingly.

1.2.4.3 Classroom and Campus Safety

I consider classroom and campus safety our shared responsibility. Please familiarize yourself with the classroom emergency placard, which contains important information that could prove useful should an emergency arise. Emergency situations could include a fire alarm, tornado warning, medical emergency, or hostile intruder, and our response might consist of evacuating the building in an orderly manner, moving as a group to a different location in the building, rendering assistance a class member, or staying put and locking the classroom door. In all cases, I will provide clear instructions as to what you should do.

Please make sure to have your primary mobile number listed in Banner so that you will automatically receive any campus alerts. Should you receive an emergency notification in class, notify me immediately.

You can find more information about campus safety can be found at the sites shown in Table 1.2.3, p. 2.

Table 1.2.3 Safety Information Locations

Information	Resource Location
Directions for enabling additional contact means to the campus alert system	`https://www.gvsu.edu/gvsualert`
List of campus tornado safety locations	`https://www.gvsu.edu/emergency/tornado-safety-location-2.htm`
Active shooter response	`https://www.gvsu.edu/dps/active-shooter-response-85.htm`
Medical emergencies and campus AED locations	`https://www.gvsu.edu/emergency/medical-emergency-15.htm`

1.3 Introduction and Teaching Philosophy

1.3.1 Introduction to Math 201

1.3.1.1 Introduction

Calculus, Math 201 is designed to give you tools to succeed in fields where deep understanding of relationships among changing quantities is needed.

1.3.1.2 Course Description

> Math 201 is a study of how changing quantities interact, and how quantifying these relationships can be applied. In order to understand these ideas, it is important to see where they arise. This means analyzing data or situations, choosing among appropriate models, and solving applied problems. Of course, we can't predict every situation you will encounter in your future, so it is also important to develop flexibility and resilience.

More information can be found in Content, p. 3 and Objectives, p. 4.

1.3.2 Teaching Philosophy

1.3.2.1 Learning is fun!

Learning is about feeding our curiosity, stretching ourselves to be better, finding out, and understanding. Those things are all rewarding, which is another word for *fun*. When learning isn't fun, there is something badly wrong.

You know what isn't fun? Grades. Getting them, assigning them, working for them — none of these things are fun. They are, in many ways, the *opposite* of fun.

A growing body of evidence shows essentially no correlation between grades and success in future endeavors. Anecdotal evidence and educational research support the idea that grades, in fact, create a *dis-incentive* for learning and create incentives for several negative behaviors, including engaging in academic dishonesty ("cheating"). Neuroscience research further shows that dopamine is critical for learning, except in the case of learning with punishment. In other words, *you need fun to learn*. See, for example, `elcenia.com/iamapirate/dopamine.pdf`

One thing we will do in this course is remove the pressure of the grade using a version of specifications grading that is sometimes called *ungrading*. More details are in Assessment, p. 7.

1.3.2.2 Math is fun!

Many of you may have an experience of mathematics that does not suggest math is fun. Even if you know and appreciate the *usefulness* of math, the subject is often presented in a way that sucks all the fun out of it. Imagine sitting down to play a video game and finding that before you can actually start playing the game, you must first spend hours and hours memorizing cheat codes! "Now we need to master the *up, up, down, down, left, right, left, right, B, A, Start* sequence and its variations. This will make later games easier. Enter this sequence 35 times on the next screen."

This is nothing like what mathematics really is. Yes, sometimes memorizing techniques can make later things easier, but these techniques have deeper meanings and origins. They are not just a random sequence of buttons to push.

Instead, mathematics is about analyzing situations, carefully describing or quantifying them, and looking for (more or less universal) patterns. That may still sound dry, but my personal goal in this class, in addition to giving you useful techniques, is to help you to see the beauty and joy of mathematics.

1.4 Objectives and Content

1.4.1 Content

1.4.1.1 Course Description

Math 201 is about understanding instantaneous rates of change: calculating rates of change numerically and analytically; understanding relationships between a function and its rate of change function; interpreting rates of change graphically, numerically, and practically; using rates of change to solve problems. Toward this end, we also introduce the concepts of limit and convergence, and use them to elucidate the concept of continuity.

1.4.1.2 Topics

Topics for this course include:

- Limits and continuity

- The concept of a derivative including the derivative at a point, the derivative of a function, the second derivative, interpretations of the derivative and second derivative, and linear approximations using derivatives
- Derivatives of basic functions including polynomial functions, exponential and logarithmic functions, trigonometric and inverse trigonometric functions
- Rules for differentiation including the product rule, the quotient rule, the chain rule, and implicit differentiation.
- Use first and second derivatives to help solve optimization problems, to study families of functions, to solve related rates problems, and to use L'Hopital's rule for evaluating limits
- The definite integral, the Fundamental Theorem of Calculus, and theorems about definite integrals

1.4.1.3 Additional Content

In addition to developing mathematical tools, techniques, and skills, we will seek to further our understanding of the role and importance of mathematics as a not only a tool, but as a way of making sense of patterns.

1.4.2 Objectives

The syllabus of record for MTH 201 states:

> After successful completion of the course, students will be able to
>
> 1. ***Application/Apply—Apply:*** Students will be able to apply the definite integral to compute the total change of a function from the rate of change and to compute area.
> 2. ***Knowledge/State—State:*** Students will know the Fundamental Theorem of Calculus.
> 3. ***Knowledge/Describe—Describe:*** Students will be able to describe the relation between the definite integral of a function and areas between the graph of that function and the horizontal axis.
> 4. ***Knowledge/Define—Define:*** Students will understand the definition of the definite integral as the limit of a Riemann sum.
> 5. ***Application/Compute—Compute:*** Students will be able to compute derivatives using the product rule, quotient rule, and chain rule.
> 6. ***Evaluation/Evaluate—Evaluate:*** Students will be able to conceptually understand and evaluate limits, including L'Hopital's Rule.
> 7. ***Knowledge/State—State:*** Students will be able to state the formal definition of the derivative of a function and the purpose of each symbol in that definition.
> 8. ***Comprehension/Discuss—Discuss:*** Students will be able to discuss the interpretations of the derivative as the slope of a tangent line and an instantaneous rate of change of a function.
> 9. ***Guided Response/Determine—Determine:*** Students will be able to determine the tangent line to a function at a specific point on the graph of the function.
> 10. ***Guided Response/Determine—Determine:*** Students will be fluent with the concept of function and be able to determine information about a function through the processes of differentiation and integration.

Furthermore, Grand Valley State University's mission statement is: "At Grand Valley State University, we empower learners in their pursuits, professions, and purpose. The university enriches society through excellent teaching, active scholarship, advancement of equity, and public service."

This course is a step along the way to you achieving your own professional and personal goals. In light of this, some of my goals for this course are for you

- to develop the skill of thinking mathematically,
- to have a stronger idea of how discrete mathematics relates to computer science, and
- to have an idea of what mathematics.

1.4.3 Materials and Resources

1.4.3.1 Primary Reference

The main text for this course will be *Active Calculus* by Matthew Boelkins, ISBN13:978-1724458322. This is available as an interactive on-line text (`ActiveCalculus.org/single/book-1.html`) It is also available as a downloadable PDF or in hard copy as a paperback. For these options, go to the Active Calculus – Single Variable (`activecalculus.org/acs/`)

We will also use the Active Calculus Student Workbook (`activecalculus.org/wp-content/uploads/2024/08/acs-activity-workb` `pdf`) You can purchase the Student Workbook in the GVSU bookstore.

1.4.3.2 Additional Materials

Any required supplementary material used will be put placed on BlackBoard, on reserve in the library, or otherwise made available to you.

1.5 Important Dates

Aug. 31	Classes Begin
Sep. 4	100% Tuition Refund Deadline (before 5:00pm)
Sep. 4	Last Day to Add, Register, or Pay (before 5:00pm)
Sep. 7	Labor Day (No Classes)
Sep. 25	Credit/No Credit Deadline (before 5:00pm)
Sep. 25	75% Tuition Refund Deadline (before 5:00pm)
Sep. 8–18	***Goal Setting Meeting***
Sep. 29	***Progress Check 1***
Oct. 12–16	Mid-term Evaluations
Oct. 13	***Progress Check 2***
Oct. 22	Mid-term Grades Available to Students
Oct. 25–27	Fall Break
Oct. 30–Nov. 12	***Progress Meeting***
Nov. 13	Withdraw with a 'W'Grade (before 5:00pm)
Nov. 25–29	Thanksgiving Day recess
Nov. 16–20	***Progress Check 3***
Dec. 12	Classes End and Commencement
Dec. 12–19	Examination Week
Dec. 14–18	***Final Meetings***
Dec. 19	Semester Ends
Dec. 22	Grades Due
Dec. 24	Grades Available to Students

Chapter 2

Course Documents

2.1 Assessment

2.1.1 Introduction to Ungrading

As mentioned earlier, substantial evidence shows that grades and grading are inherently antithetical to learning. *Ungrading* shifts the goal of a course from achievement to learning.

It is also important to note that the content of this course focuses on developing conceptual and procedural fluency with a set of concepts and tools as well as developing a "mathematical mindset". Thus, it makes sense to have learning targets that students need to satisfy in order to consider the course successful.

Ultimately, though, this is your course. It is you who are the learner, and so you are in the best position to determine what will be of the most interest, what will be the most engaging, and, beyond the basics, what will be of the most value to you. More than that, *you* are the best person to determine how much you have learned. Therefore, you will be an active participant in determining the direction of the course and in your evaluation.

To put things bluntly, I will determine, based on a set of learning targets and a body of evidence that you present to me, whether you have done the work necessary to pass the class. Beyond that, *you* will determine your own grade, based on the evidence you present that you have met your own goals, as outlined in Portfolio and Reader's Guide, p. 7, Portfolio and Reader's Guide, p. 7.

2.1.2 Portfolio and Reader's Guide

During the first week of class, we will discuss as a class what success in this class might look like, and then you will meet individually with me to develop a set of personalized goals. (See Important Dates, p. 5.)

These goals will not about grades and grading. Instead, you are invited to consider your *learning*, and more generally what *success* in this course will be for you.

Over the course of the semester, you will assemble a comprehensive portfolio that demonstrates that you have met the basic requirements for the course and demonstrates your progress toward your goals. Part of this portfolio will be three *Progress Checks*.

In addition to the Progress Checks, you are expected to regularly submit items to me for focused feedback. Based on this feedback, you will reflect, rework, extend, and revise as necessary to meet basic requirements and show forward progress.

At least one time, mid-semester, you will schedule an individual meeting to discuss your progress, your successes, and any challenges you are facing.

For each individual meeting, you will prepare and share with me an agenda or guide ahead of time with specific items you wish to discuss. This guide should discuss successes, challenges, and progress, and should direct the reader where to find evidence of these items. For example, you might include something along the lines of "*Successes* Since our last meeting, I have reviewed and improved my understanding of difference quotients and average rate of change. Some good examples of my understanding and interpretation can be found in *Driving North/South* and *Filling a Tank*."[1]

[1]These titles are completely invented and may not be anything like the titles of activities we actually use.

Your guide should also address the following questions.

- What are you particularly proud of or excited about?
- What has been challenging?
- What do you plan to do next?, and
- What do you need from me (the instructor) to facilitate your progress toward your goals?

For your final portfolio conference, which will be during the week of final exams, you will submit your final portfolio (with a final version of your reader's guide), again citing items of interest. Then, based on your learning and progress toward your goals and your submitted evidence, you will tell me what grade you have earned for the course.

2.1.3 Required Elements of the Portfolio

The following items are required to be in your portfolio.

2.1.3.1 Reader's Guide and Progress Report

Depending on how you choose to organize your portfolio, this could simply be a table of contents plus a brief discussion of your progress.

For each submission of your portfolio, you will update your report to discuss your overall progress, to highlight progress since the last check-in, and to cite specific items of interest.

The goal of this is to help the reader (your instructor) prepare for the meeting with you and to make items easy to find.

One way to think of this is to think of answering the questions,

- What are you particularly proud of or excited about?
- What has been challenging?
- What do you plan to do next?, and
- What do you need from me, the instructor, to facilitate your progress toward your goals?

2.1.3.2 Progress Checks

Several times throughout the semester, we will have in-class progress checks. These are opportunities for you to show your understanding of concepts and fluency with calculations.

When your progress checks are returned to you, you are expected to put them in your portfolio and write a reflection on your performance, noting things that went well as well as areas of future growth.

If there were substantial errors or misconceptions, those should also be addressed in the reflection.

2.1.3.3 Catalog of Detailed Examples

Over the course of the semester we will be study functions and their derivatives from graphical, analytic, numerical, and conceptual viewpoints. For each topic, whenever possible, you must include at least one detailed example of the use of that concept in a practical setting. That is, each example must:

- Begin with a practical situation where the numbers have units and meaning.
- Include a discussion of why the particular concept or tool was used (possibly including why other approaches would not be appropriate).
- Show the process of using the concept to understand and solve the problem.
- Include multiple representations, such as graphical, numerical, and algebraic representations.
- Interpret the solution in context, giving a result with practical implications.

2.1.3.4 Overviews of Activities, Techniques, and Applications

Over the course of the semester you will be assigned multiple activities, investigations, and projects dealing with techniques, concepts, and applications of calculus.

For each *topic* we study, not each activity, you are expected to demonstrate understanding of the main mathematical ideas and an ability to apply any associated techniques.

Like everything in the portfolio, these are works in progress, and you are expected to revisit them in light of later activities or readings that may change your understanding of them.

2.1.3.5 Citation of Sources

If you make use of any outside resources for this class, you are required to cite those sources. The particular format of citations is not important. If you need specificity, please follow MLA standards.

2.1.4 Optional Elements of the Portfolio

You can (and are encouraged to) include any items in your portfolio that demonstrate progress toward your individual goals. The following items are some things that previous students have chosen to include.

2.1.4.1 Details of Activities and Investigations

In addition to the basics for an activity, you may choose to include more.

For instance, you might include work to show that you have:

- gained additional depth of understanding of important concepts
- understood the historical or philosophical importance of a topic (existence of solutions versus calculation of solutions), or
- made connections to other topics in this course, in other courses, your major, or to everyday life.

2.1.4.2 Experiments and Explorations

If there is a topic from class that you feel does not go far enough, you are free to investigate more deeply. For example, if you particularly enjoy global optimization, you might look at a more complex optimization, such as minimizing the construction cost for an oil barrel.

2.1.4.3 Summaries of Outside Reading

If you use outside sources to guide your learning, you may want to include more than just citations. You can summarize what you learned from those sources and include the summaries.

2.1.4.4 Other Items

Other items may be included in your Portfolio if you they demonstrate your progress in meeting your course goals. Some examples of what you may choose to include are practice problems, conjectures, or expository papers.

2.1.5 Professionalism

For this course, I define professionalism to mean respect for yourself, your classmates, and your instructor. This includes (but is not limited to)

- consistent forward progress and timely completion of assignments,
- prompt scheduling of and on-time arrival for individual meetings,
- preparation for class, and
- attendance and participation in class.

A pattern of unprofessional behavior during the semester may, at my discretion, result in a course grade penalty of up to two full letter grades (20%).

2.1.6 Culminating Experience

There will be no final exam in this course. Instead, your individual conference with me during the final period will serve as the culminating experience in this course.

2.2 Course Policies and Expectations

This course is subject to the GVSU policies listed at `http://www.gvsu.edu/coursepolicies/`. The following policies also apply.

2.2.1 Use of Generative AI

Use of Generative AI Large Language Models [LLMs] (such as ChatGPT, Gemini, Copilot, etc.) in this class is subject to a few conditions, namely Awareness, p. 10, Transparency, p. 10, and Accountability, p. 11. Transparency and accountability are fairly straight-forward, and are closely related to Academic Integrity, p. 13, but awareness is both more subtle and arguably more important.

2.2.1.1 Awareness

If you plan to use generative AI in this class, be aware of what you are using it for and whether it helps you toward your goals or just wastes your time. I find a useful analogy to be the idea of ***taking a forklift to the gym***.

If your goal in going to the gym is to strengthen your muscles by lifting weights, using a forklift to lift the weights defeats the purpose. Sure, you can get in your reps, you can "lift" more than you could without the forklift, and you can do more reps and not get tired, but it doesn't work toward your goal of strengthening your muscles. It is a waste of your time. Worse, it gets in other people's way as they try to reach their strength training goals.

On the other hand, maybe you went to the gym because you work there and the boss wants you to move several large, heavy machines from one part of the building to another. If that's the case, using the forklift makes a lot of sense, because it is in fact helping with your goal of moving the machines.

Education is about strengthening your mental muscles. Be aware of how the use of heavy machinery either helps or hinders your progress.

Awareness also includes awareness of the ethical and practical pitfalls involved in using generative AI. You should be aware that

- AI is trained on the works of other people
- Plausibility, not objective truth, is the training goal of AI. It often makes things up that sound plausible.
- AI is resource intensive. For example, according to `what-uses-more.com`, a paragraph of AI generated text uses more than 23 times as much electricity and water than a Google search without AI.

2.2.1.2 Transparency

When you use generative AI, be transparent: make note of exactly what you asked the AI to do (your prompts), and exactly what part of the AI results you used in your materials. For example, you might include something like the following in your portfolio.

> "In this problem, we were asked to model the number of hours of daylight as a function of the date. I knew this would be either sine or cosine, but I still didn't understand how to make the right days line up with the peaks and valleys, so I asked Copilot to explain. My prompt was 'Explain how to shift a sine function so that the peaks and valleys line up with data.'"
>
> "Copilot started by giving a formula for a sine function as $y = A\sin\big(B(x - C)\big) + D$ and then gave step-by-step alignment guide. The important part for me was how to adjust the 'phase shift', C. It said to 'Find where a peak (or valley) occurs in your data, and shift the sine wave so its peak matches that point: $C = x$-value of desired peak.' I was then able to align the peak amount of sunlight with June 21 by using $C = 172$ days since the beginning of the year."

2.2.1.3 Accountability

When you use generative AI, you are accountable. What that means is that you must be able to explain all of your work and understand the power and the limitations of using LLMs.

Using the example of fitting a sine wave, a student submitting this should have noticed that ***Copilot got it wrong***. Setting $C = 172$ places the *midline* of the sine function at $x = 172$ days. Using $C = 172$ in the formula $y = A\cos\big(B(x - C)\big) + D$ would place the peak at $x = 172$, or you could find the date of the spring equinox and set C to that value in $y = A\sin\big(B(x - C)\big) + D$.

2.2.2 Initial Drafts of Work in Timed Settings

Initial drafts of work done in class (a setting with limited time) are not expected to be typeset. Instead, high resolution images or digital scans of your initial work can be included in your portfolio.

Reflections on your work, revisions, and corrections, though, are expected to follow the guidelines given for work in un-timed settings.

2.2.3 Submitting Un-Timed Work for Feedback

2.2.3.1 Format of Submissions

All drafts of work done in settings without prescribed time constraints, such as exercises, proofs, expository writing, and so forth,

- must be written using the software capable of high-quality mathematical typesetting or neatly hand-written and scanned;
- must be spell-checked and proofread for grammatical correctness;
- must be submitted electronically, via BlackBoard or shared link; and
- must be available in a form suitable for commenting and editing.

See Mathematical Typesetting, p. 14, for more information about options for mathematical typesetting software.

2.2.3.2 FERPA and Email

Please note that due to security issues and based on guidance from the Department of Education regarding FERPA, the Family Educational Rights and Privacy Act of 1974, I cannot use e-mail or any unsecured platform to discuss grades.

2.2.4 Time Management

One of my goals for this course is to challenge you to grow intellectually while giving you the tools and environment that will allow you to succeed. If you at any time feel that you are not being challenged to achieve your fullest potential in understanding the concepts we address, or if you have comments about the pace of the class or any other concerns, I encourage you to discuss your concerns with me during my student hours, p. 1 or an appointment with me.

As a university mathematics course, you should expect the course to be challenging. While we will all work to make time spent in class as productive as possible, it is not possible to learn everything using class time alone: much of your learning and understanding will occur as you reflect on what we have done in class, work on problems, and read and discuss course material outside of class.

The most important rule for this course is *do not get behind in your work.* Using a portfolio removes the inherent penalties in a slow start, but the trade off is that it can be tempting to put things off until later. Always stay up to date with the material in this course. Make succeeding in this course a priority!

2.2.5 Getting More from Class

Learning cannot take place solely in the classroom, and you should expect to put in considerable work outside of class. Different students learn in different ways and require different types of support. I am committed to providing you with the support that you require, whether that means being a critical questioner of your processes and results, providing guidance and mentoring, or something else entirely.

This section has some advice and perspectives that I hope will help you make the most of your time and effort in this class.

2.2.5.1 Attend Student Hours

First, never hesitate to come to me during student hours, p. 1 or to make an appointment to see me. My primary job is to facilitate your understanding and learning, so when you come to me you are allowing me to do my job well.

2.2.5.2 Form a Study Group

You may also find that working with a group of classmates is beneficial. I encourage you to form groups to work together outside of class. Working with a group also allows you to see things from different perspectives.

If you or your study group have questions, please come to me and we will work through them together. If my student hours, p. 1 are at times that you cannot make due to other commitments, make an appointment (`mailto:wellsc@gvsu.edu`) to see me at another time.

2.2.5.3 Make Attendance a Priority

As discussed in Assessment, p. 7, attendance in class is an important part of this course. You are expected to be present and on time each day we meet. You are responsible for announcements made in class concerning material covered, assignments, changes in the syllabus or due dates, or anything else pertinent to the course. Furthermore, many of the activities from class and reflections on class will become part of your portfolio.

Nevertheless, your health and safety and that of your classmates is important. Therefore *if you are ill, stay at home*. In the event of a University recognized absence, arrangements can be made for you to mitigate the effects of missing class.

2.2.5.4 Come to Class Prepared

It is imperative that you work on a consistent basis. This applies to both the day-to-day work to prepare for class as well as the more long-term work such as the portfolio.

You should keep a well-organized record of your study notes, completed problems, and problems in progress for future reference. Note that these are acceptable as part of your portfolio if you believe they help make your case!

You must understand that a great deal of your learning in this course must occur on your own. It is your responsibility to complete assigned reading, read assignments carefully, complete the problems, be prepared for class, and to seek support as needed.

2.2.5.5 Participate in Class

For many people, their experience has taught them that asking questions or questioning results is frowned on as either disrespectful or inattentive. In this class, I want to cultivate an atmosphere where questioning processes, results, and reasoning is seen as showing respect that the other person's ideas are worth thinking about. This type of respectful questioning (including questioning your own processes, thoughts, and results) is crucial to becoming an effective and flexible thinker.

Asking and answering questions are not the only ways, or even the best ways, to participate in class. While we may think of class participation in terms of speaking up in class, participation can take many forms, such as active listening in groups, writing up or journaling individual or group assignments, drawing graphs or creating visual representations, sharing strategies, questioning results, and many others. In every class meeting, there will be ample opportunity for you to actively participate.

2.2.5.6 Make Full Use of Additional Learning Resources

The idea of "asking for help" can sometimes take on a negative connotation. I prefer to think of it as one way to make use of all of your available resources. If you need to travel from the Allendale campus to the Grand Rapids, you probably don't think that taking the Laker Line instead driving a car is a bad thing, it's simply another effective way to travel. And there are also lots of reasons why some students might not have access to a car, including having a flat tire or not having a license. In a similar way, there are lots of reasons to seek out additional learning resources.

In addition to student hours and individual meetings with me, GVSU provides a variety of tutoring resources, including the Math Center (`gvsu.edu/tutoring/math`) (located in Mackinac Hall). More information about other tutoring services can be found on the GVSU Tutoring (`gvsu.edu/tutoring`) website.

Using these resources is *smart*, and can be beneficial to every student.

2.2.6 Academic Integrity

2.2.6.1 Integrity of Scholarship and Grades

Truth and honesty: The principles of truth and honesty are recognized as fundamental to a community of teachers and scholars. The university expects that both faculty members and students will honor these principles and in so doing protect the validity of university grades. This means that all academic work will be done by the student to whom it is assigned without unauthorized aid of any kind. Instructors, for their part, will exercise care in the planning and supervision of academic work, so that honest effort will be positively encouraged. Compliance shall include compliance with the following specific rules:

1. No student shall knowingly, without authorization, procure, provide, or accept any materials which contain questions or answers to any examination or assignment.
2. No student shall, without authorization, complete, in part or in total, any examination or assignment for another person.
3. No student shall, without authorization, allow any examination or assignment to be completed, in part or in total, by another person.
4. No student shall knowingly plagiarize or copy the work of another person and submit it as his or her own.
5. No student shall submit work that has been previously graded or is being submitted concurrently to more than one course without authorization from the instructor(s) of the class(es) to which the student wishes to submit it.

2.2.6.2 Plagiarism

Any ideas or material taken from another source for either written or oral presentation must be fully acknowledged. Offering the work of someone else as one's own is plagiarism. The language or ideas taken from another may range from isolated formulas, sentences, or paragraphs to entire articles copied from books, periodicals, speeches, or the writing of other students. The offering of materials assembled or collected by others in the form of projects or collections without acknowledgment also is considered plagiarism. Any student who fails to give credit in written or oral work for the ideas or materials that have been taken from another is guilty of plagiarism.

Such activity may result in failure of a specific assignment, an entire course, or, if flagrant, dismissal from Grand Valley. For further information see Student Rights and Responsibilities.

2.2.7 Communication and Technology Use

2.2.7.1 Technology Requirements for Online/Hybrid Courses

Although this class face-to-face, there is always the possibility that we may have to move online (synchronous). Thus, GVSU's Technology Requirements for Online/Hybrid Courses (`gvsu.edu/online/technology-requirements-for-onlinehybrid-cour` `htm`) apply.

2.2.7.2 BlackBoard

Materials and information for this course will be posted to the course BlackBoard (`lms.gvsu.edu`) page. If you do not have access to the student network or need help in using BlackBoard, please let me know as soon as possible.

2.2.7.3 E-mail

Please check that the e-mail associated with your BlackBoard account is one that you check, as I will use that e-mail to send messages to the entire class.

2.2.7.4 WeBWorK

Practice problems will be available to you via the WeBWorK (`webwork-math.gvsu.edu/webwork2`) homework system. You are encouraged to make use of WeBWorK to practice and check your understanding. Although the feedback from WeBWorK is not as detailed as that from the instructor, it has the advantage of being essentially instantaneous!

2.2.7.5 Computer Algebra Systems

We will often make use of Desmos Graphing Calculator (`desmos.com`) and Amplify (`student.amplify.com`).

We will also sometimes use iPython interactive notebooks (such as Google Colaboratory (`colab.research.google.com`)). It is an interface to the python programming language, but ***you are not expected to learn to program***.

2.2.7.6 Mathematical Typesetting

All material submitted for your portfolio is expected to be written using using the software capable of high-quality mathematical typesetting.

Options for mathematical typesetting include markdown notebooks, such as Google Colab (`colab.research.google.com`) or Jupyter (`jupyter.org`) Notebooks, or Google Docs (`docs.google.com`).

2.2.7.7 Recording of Classes

In the event that we meet virtually via Zoom, class meetings will be recorded and will be available to students enrolled in this course as a study aid. Students who do not want their likeness during class participation included in the recorded class session may elect to turn off the camera and change their name in the Zoom window. Recordings will be available for viewing subject to the following:

- Only students enrolled in the class may view the recording.
- Students may not post or use the recordings in any other setting (e.g., social media) for any purpose. Students who violate this will be subject to student discipline, up to and including expulsion.
- If faculty have any plan to use the recording for a different class in the future, the faculty member will need written FERPA consent from those students in the current class who are identifiable in any of the recordings. A FERPA consent form signed by all students in the course will also be needed if the recordings are made available to others beyond the classroom on a non-secure digital platform.

Chapter 3

In-Class Activities

These are copies of the in-class activities distributed during the semester.

3.1 Functions and Modeling

3.1.1 Objective

The learner will practice problem solving and modeling skills, discuss strategies and approaches, and begin to appreciate the importance of modeling and functions in problem solving.

We will begin by considering the questions on our own, then move to small group discussion and finally whole class discussion.

3.1.2 Activity

Each of the questions below is deliberately ill-posed. That is, there is not enough information to answer the question. Your goal, then, is to determine what information you would need, and based on that information, make a plan to find the answer. (Note: you cannot say the information you need is the answer!)

1. How deep is the water?
2. How much water is in the tank (as in, how many glasses of water you could fill)?
3. How long would it take for the water level to rise 3cm?
4. How much would the water level change if you added 3L of water?
5. How long would it take to fill the tank?

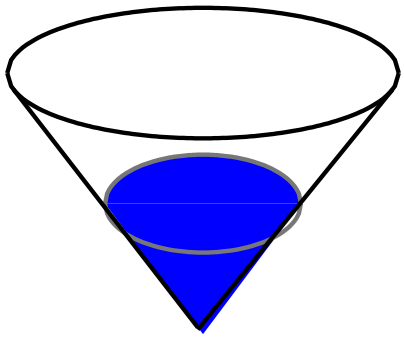


Figure 3.1.1 A Conical Tank Partially Filled with Water.

3.2 Understanding Limits

Goal: The learner will use graphical, numerical, and/or analytical means to calculate and understand the process of calculating the value of a limit and will write a careful explanation of their thinking and calculations.

3.2.1 Activity

Instructions: The questions below are intended to guide your understanding as you investigate two very similar functions. Work together to answer the questions. Then

- Create a drawing, graph, or visual representation that helps explain your findings.
- Write a brief guide for your process, results, and understanding of the results.
- For the studied functions and points, either write ``the function has a limit at the point'' or ``the function does not have a limit at the point''.
- For each point, decide if you think f or g is ``nicer'' at the point (or if they are equally nice), and write an explanation of your decision.

1. Let the function f be given by
$$f(x) = \begin{cases} \frac{x^3-8x^2+17x-10}{x-2}, & x < 2 \\ 4, & x = 2 \\ \frac{-2x^3+22x^2-76x+80}{x-2}, & x > 2. \end{cases}$$
   (a) Find $f(3)$ or explain why it is not defined.
   (b) Find $\lim_{x\to 3} f(x)$ or explain why it is not defined.
   (c) Find $f(2)$ or explain why it is not defined.
   (d) Find $\lim_{x\to 2} f(x)$ or explain why it is not defined.
   (e) Find $f(4)$ or explain why it is not defined.

(f) Find $\lim_{x\to 4} f(x)$ or explain why it is not defined.

2. Let the function g be given by

$$g(x) = \begin{cases} \frac{x^3-8x^2+17x-10}{x-2}, & x < 2 \\ \frac{x+2}{x-3}, & x = 2 \\ \frac{-2x^3+22x^2-67x+62}{x-2}, & x > 2. \end{cases}$$

(a) Find $g(3)$ or explain why it is not defined.

(b) Find $\lim_{x\to 3} g(x)$ or explain why it is not defined.

(c) Find $g(2)$ or explain why it is not defined.

(d) Find $\lim_{x\to 2} g(x)$ or explain why it is not defined.

(e) Find $g(4)$ or explain why it is not defined.

(f) Find $\lim_{x\to 4} g(x)$ or explain why it is not defined.

3. Let the function k be given by

$$k(x) = \begin{cases} \frac{x^3-8x^2+17x-10}{x-2}, & x < 2 \\ -3, & x = 2 \\ \frac{-2x^3+22x^2-67x+62}{x-2}, & x > 2. \end{cases}$$

(a) Find $k(3)$ or explain why it is not defined.

(b) Find $\lim_{x\to 3} k(x)$ or explain why it is not defined.

(c) Find $k(2)$ or explain why it is not defined.

(d) Find $\lim_{x\to 2} k(x)$ or explain why it is not defined.

(e) Find $k(4)$ or explain why it is not defined.

(f) Find $\lim_{x\to 4} k(x)$ or explain why it is not defined.

3.3 Rate of Change of Depth

Objectives

- In this activity, we will revisit the conical water tank with extra information and seek to find out how quickly the water in the tank is rising at various times.

  (Technically, the shape of the tank is a *right circular cone*, which means that cross-sections are perpendicular to the vertical axis and are circles.)

We are considering a (hypothetical) conical water tank, oriented with the point downward. We also have the following information.

- The tank has a height/depth of 6m.
- The radius of the tank at the top is 2m.
- Water is running into the tank at a steady rate of 20πl/s.
- The tank has been filling before we begin measuring.
- At the time we start measuring, the tank is filled with water to a depth of 2m.

1. Recall that the volume of a right circular cone is given by $V = \frac{1}{3}\pi r^2 h$, where r is the radius and h is the vertical height.

(a) Find the total volume of the tank (in cubic meters).

(b) Express the volume of the cone in liters.

2. We will now find a formula for the volume of water in the tank as a function of time.

(a) Recall that we can use similar triangles to find the relationships of the total height of the tank, the radius at the top of the tank, any depth of water, and the radius of the water at that depth.

Find the amount of water in the tank when we begin measuring. That is, when the height/depth of the water is 2m. This is the volume at *time* $t = 0$.

(b) Use the volume at time $t = 0$ and the rate at which water is running into the tank to express the volume *as a function of the time* t (in seconds) since beginning to measure.

Your expression should have the form $V = F(t)$, where V is expressed in liters and t is expressed in seconds.

(c) Given the initial volume of water, the total volume of the tank, and the fill rate, determine how long (after time $t = 0$) it will take for the tank to be completely full.

3. Sketch a graph showing the depth of water over time. To do this, you will want to consider:

- how deep the water is at time $t = 0$;
- how deep the water is when the tank is full;
- how long it takes for the tank to be completely full; and
- whether the depth rises at a steady rate, rises quickly at first and then more slowly, or rises more slowly at first and then more quickly.

4. We will now express the volume *as a function of depth*.

(a) Let d represent the depth of water as measured from the bottom of the tank. Use similar triangles to express the radius at a given depth in terms of d.

(b) Use the formula for the volume of a cone and your expression for the radius at a given depth to find the volume of (the cone of) water in the tank as a function of d.

Your expression should have the form $V = G(d)$ where V is expressed in liters (convert it if your formula is for cubic meters!) d (in meters) is the only variable on the right hand side.

5. Now we will combine our results to obtain a function for the depth, d, as a function of time t.

(a) Set the volumes from Worksheet Exercise 3.3.2, p. 18 and Worksheet Exercise 3.3.4, p. 19 equal to each other. That

is, write $F(t) = G(d)$.

(b) Solve your equation for d. This gives you a function $f(t)$ so that $d = f(t)$.

(c) Use appropriate technology (such as Desmos (`desmos.com`) or a graphing calculator) to graph your function. How does it compare to your sketch in Worksheet Exercise 3.3.3, p. 19?

6. Our goal now is to approximate the rate at which the water is rising at various times. We will use *average rates* over small intervals to approximate *instantaneous rates*. Then, in our next activity, we will investigate a useful algebraic identity and use it to calculate the instantaneous rates *exactly* based on our function.

Note: for the rate at $t = 0$, we will use the fact that *the tank was filling before we started measuring*, and thus (small) *negative times* are valid.

Note the approximations in this exercise use what is called a *central difference quotient* because we are using times an equal amount before/left of the point of interest and after/right of the point of interest.

More generally, a central difference quotient for a function f about a point of interest $t = a$ will have the form $\frac{f(a+h)-f(a-h)}{(a+h)-(a-h)} = \frac{f(a+h)-f(a-h)}{2h}$

For numerical approximations of the instantaneous rate, using central difference quotients turns out to be more accurate than using difference quotients between the point of interest and another point. In the limit as the differences approach zero, they yield the same result.

(a) We will approximate the average rate of change of depth at $t = 5$.

(i) Use your function to calculate the average rate of change in depth between $t = 4.5 = 5 - 0.5$ and $t = 5.5 = 5 + 0.5$. (This calculation is the *central difference quotient* about $t = 5$ with $h = 0.5$.)

(ii) Use your function to calculate the average rate of change in depth between $t = 4.9 = 5 - 0.1$ and $t = 5.1 = 5 + 0.1$. (This calculation is the *central difference quotient* about $t = 5$ with $h = 0.1$.)

(iii) Use your function to calculate the average rate of change in depth using central difference quotient about $t = 5$ with $h = 0.01$.

(b) Repeat Task 3.3.6.a.i, p. 20 with central differences around $t = 2$.

(c) Repeat Task 3.3.6.a.i, p. 20 with central differences around $t = 0$. ***Note that this uses negative values for*** t***, which is completely valid.***

3.4 Instantaneous Rate of Change of Depth

Objectives

- In this activity, we will continue using the conical water tank as we develop our understanding of instantaneous rates of change.

The first few exercises in this activity will investigate a useful algebraic pattern/identity that we will use fairly often throughout the course. Then, we will return to the function we developed in Worksheet 3.3, p. 18 and use the algebraic identity to calculate the instantaneous rate of change of depth at various times.

1. Many of us are aware of the algebraic identity used for factoring a difference of squares, $(x^2 - y^2) = (x - y)(x + y)$, but you may not be aware that it is a special case of a much more general pattern.

You may find it useful in the following exercises to use an "area model" array for multiplying polynomials. For example, to multiply $(x - y)(x^2 + 3xy + y^2)$, I could use a model as follows:

	x^2	$+3xy$	$+y^2$
x	x^3	$+3x^2y$	$+xy^2$
$-y$	$-x^2y$	$-3xy^2$	$-y^3$

Collecting like terms (which lie along diagonals), we see that $(x - y)(x^2 + 3xy + y^2) = x^3 + 2x^2y - 2xy^2 - y^3$.

(a) Calculate $(x - y)(x^2 + xy + y^2)$.

(b) Calculate $(x - y)(x^3 + x^2y + xy^2 + y^3)$.

(c) Calculate $(x - y)(x^4 + x^3y + x^2y^2 + xy^3 + y^4)$.

(d) Let n be some (arbitrary) integer (greater than 4). Based on your previous results, what do you think would be the result of calculating $(x - y)(x^n + x^{n-1}y + x^{n-2}y^2 + \cdots + xy^{n-1} + y^n)$?

(e) Use your previous results, making appropriate substitutions, to calculate $\left(b^{1/3} - a^{1/3}\right)\left(b^{2/3} + b^{1/3}a^{1/3} + a^{2/3}\right) = \left(b^{1/3} - a^{1/3}\right)\left((b^{1/3})^2 + (b^{1/3})(a^{1/3}) + (a^{1/3})^2\right)$.

(f) Based on previous results, what could you multiply $b^{1/4} - a^{1/4}$ by in order to obtain $b - a$ as the product?

(g) Use your previous results, making appropriate substitutions, to calculate $\left((kb+c)^{1/3} - (ka+c)^{1/3}\right)\left((kb+c)^{2/3} + (kb+c)^{1/3}(ka+c)^{1/3} + (ka+c)^{2/3}\right)$.

2. In the exercise, we will make use of the algebraic techniques we have developed to calculate the instantaneous rate of change for the function $f(t) = \sqrt[3]{0.54t+8} = (0.54t+8)^{1/3}$

(a) Write the difference quotient $\frac{f(b)-f(5)}{b-5}$.

(b) Find an appropriate expression so that

$$\begin{aligned}\left((0.54b+8)^{1/3} - (0.54\cdot 5+8)^{1/3}\right)(\ldots?\ \ldots) &= (0.54b+8) - (0.54\cdot 5+8)\\ &= 0.54(b-5)\end{aligned}$$

(c) Multiply the numerator and denominator of the difference quotient in Task 3.4.2.a, p. 23 by the expression you found in Task 3.4.2.b, p. 23 and simplify the numerator.

(d) Note that for any $b \neq 5$, $(b-5) \neq 0$, and so $\frac{(b-5)}{(b-5)} = 1$. Therefore, for every b except $b = 5$, your expression in Task 3.4.2.c, p. 23 is identical to the same expression with the $(b-5)$ in the numerator canceled by the $(b-5)$ in the denominator, and therefore their limits as $b \to 5$ are identical.

However, the expression with $(b-5)$ canceled is continuous at $b = 5$ and so its limit as $b \to 5$ is simply its value *at* $b = 5$.

Evaluate this limit, and thus calculate

$$\lim_{b\to 5} \frac{f(b)-f(5)}{b-5}.$$

This is the *instantaneous rate of change* for f at $t = 5$.

3. Repeat Worksheet Exercise 3.4.2, p. 23 for $\frac{f(b)-f(2)}{b-2}$ to find $\lim_{b\to 2} \frac{f(b)-f(2)}{b-2}$.

4. Repeat Worksheet Exercise 3.4.2, p. 23 for $\frac{f(b)-f(0)}{b-0}$ to find $\lim_{b\to 0} \frac{f(b)-f(0)}{b-0}$.

5. Repeat Worksheet Exercise 3.4.2, p. 23 for $\frac{f(b)-f(a)}{b-a}$ to find $\lim_{b\to a} \frac{f(b)-f(a)}{b-a}$, where a is an unspecified value.

3.5 Interpretations of the Derivative

Objectives

- The learner will use context and units to interpret the meaning of the derivative in practical settings and use the derivative at a point to estimate the value of the (original) function at nearby points.

Recall that for a function $f(t)$, the derivative at the point $t = a$ is defined by

$$f'(a) = \lim_{b \to a} \frac{f(b) - f(a)}{b - a}$$

or equivalently

$$\begin{aligned} f'(a) &= \lim_{h \to 0} \frac{f(a+h) - f(a)}{(a+h) - a} \\ &= \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}. \end{aligned}$$

Noting that $f(b) - f(a) = \Delta f$ is the change in output and $b - a = \Delta t$ is the change in the input variable, we can think of the derivative as

$$\begin{aligned} f'(a) &= \lim_{\Delta t \to 0} \frac{\Delta f}{\Delta t} \\ &= \frac{\mathrm{d}f}{\mathrm{d}t}, \end{aligned}$$

and the *units* of the derivative will be the units of the output of the function divided by the units of the input to the function.

For example, with the function $s(t) = \left(64 - 16(t-1)^2\right)$ ft that takes time t in seconds as input and gives height in feet as its output, we have

$$\begin{aligned} s'(a) &= \lim_{b \to a} \frac{(s(b) - s(a)) \text{ ft}}{(b - a) \text{ s}} \\ &= (-32a + 32) \text{ ft/s}. \end{aligned}$$

We also know that we can use numerical approximations, so for a given (small, finite) value of Δt,

$$\begin{aligned} f'(a) &\approx \frac{\Delta f}{\Delta t} \\ &= \frac{f(a + \Delta t) - f(a)}{\Delta t} \\ &= \frac{f(b) - f(a)}{b - a}. \end{aligned}$$

But, because $f'(a) \approx \frac{f(b)-f(a)}{b-a}$, we also have $f(b) - f(a) \approx f'(a)(b - a)$. In other words, we can approximate the change in f from the change in t along with the derivative!

In the following scenarios, use the context and the units to help you interpret the meaning of the derivative of the appropriate functions, and use the derivative to estimate the value of the original function at nearby points, as appropriate.

1. Production Costs. The Bedrock Gravel Company has hired a consulting firm to help them manage gravel production costs. As a first step, the company uses information on past production to model monthly production costs. They develop a function, C, which gives the total monthly production cost (in dollars) as a function of the number of tons, T, of gravel produced.

Below are some questions about some numerical findings. For each, pay attention to units, and keep in mind the practical (that is, economic) interpretations, *including appropriate units.*

(a) Consider the equation $C(324) = 15,770$.

(i) What are the units of 324?

(ii) What are the units of $15,770$?

(iii) What is the economic interpretation of $C(324) = 15,770$?

(b) Now consider the equation $\dfrac{C(225)}{225} = 59.89$.

(i) What are the units of 225?

(ii) What are the units of $C(225)$?

(iii) What is the economic interpretation of $\dfrac{C(225)}{225} = 59.89$?

(c) Next, consider the equation $\dfrac{C(324) - C(225)}{324 - 225} = 23.18$.

(i) What are the units of $C(324) - C(225)$?

(ii) What are the units of $324 - 225$?

(iii) What is the economic interpretation of $\dfrac{C(324) - C(225)}{324 - 225} = 23.18$?

(d) Recall that $C'(324) = \lim_{b \to 324} \dfrac{C(b) - C(324)}{b - 324}$.

(i) What are the units of $C'(324)$?

(ii) What is the economic interpretation of $C'(324) = 21.25$?

2. **Snowfall.** On one particular morning, consider the function S which takes as input the number of hours since midnight and give as its output the number of inches of snow that have fallen between midnight and the given time. Assign appropriate units to each of the quantities listed and answer the question.

(a) What is the practical meaning of $S(3) = 8$?

(b) What is the practical meaning of $S'(3) = 2$?

(c) Assuming $S(3) = 8$ and $S'(3) = 2$, make a prediction for the value of $S(3.25)$?

3. Energy Storage. An energy company pumps water into large holding tanks during times of low demand to store energy to use for electricity generation in times of high demand. The function $E(h)$ gives amount of energy stored (in Watt-hours) as a function of the height of the water column (that is, the depth), h, of water in the tank (in meters).

(a) Including appropriate units, explain the practical meaning of $E(3.00) = 5.935$.

(b) Including appropriate units, explain the practical meaning of $E'(3.00) = 4.320$.

(c) Assuming $E(3.00) = 5.935$ and $E'(3.00) = 4.320$, estimate and interpret

(i) $E(3.04)$

(ii) $E(3.08)$

(iii) $E(2.97)$

(d) Find an expression that would allow you to approximate $E(h)$ for any value of h that is close to 3.00.

Hint. Take note of the steps you use to answer the questions in Task 3.5.3.c, p. 28 and use that algorithm to develop a function that records those calculation steps.

3.6 Sketching the Derivative Function

Objectives

- The Learner will develop graphical understanding of the instantaneous rate of change of a function.
- The Learner will develop numerical understanding of the instantaneous rate of change of a function.

Figure 3.6.1, p. 29 shows the graph $y = f(x)$, along with the tangent line to the graph at $(a, f(a))$ for $a = 10$.

The goal of this activity is to sketch the graph of the *instantaneous rate of change* of f by plotting points.

Notice that the tangent line at $\big(10, f(10)\big)$ goes through points at approximately $(5, 17.2)$ and $(15, 7.4)$. Thus, we can estimate the slope of the tangent line by $\frac{17.2-2.4}{5-15} \approx -0.98$.

Remember that graphically we can interpret the instantaneous rate of change of a function at a point as the slope of the tangent line at that point. This means that we know that a point on the graph of the instantaneous rate of change is approximately $(10, -0.98)$.

We have plotted the point $(10, f'(10) \approx (10, -0.98)$, one point on the graph of $f'(x)$.

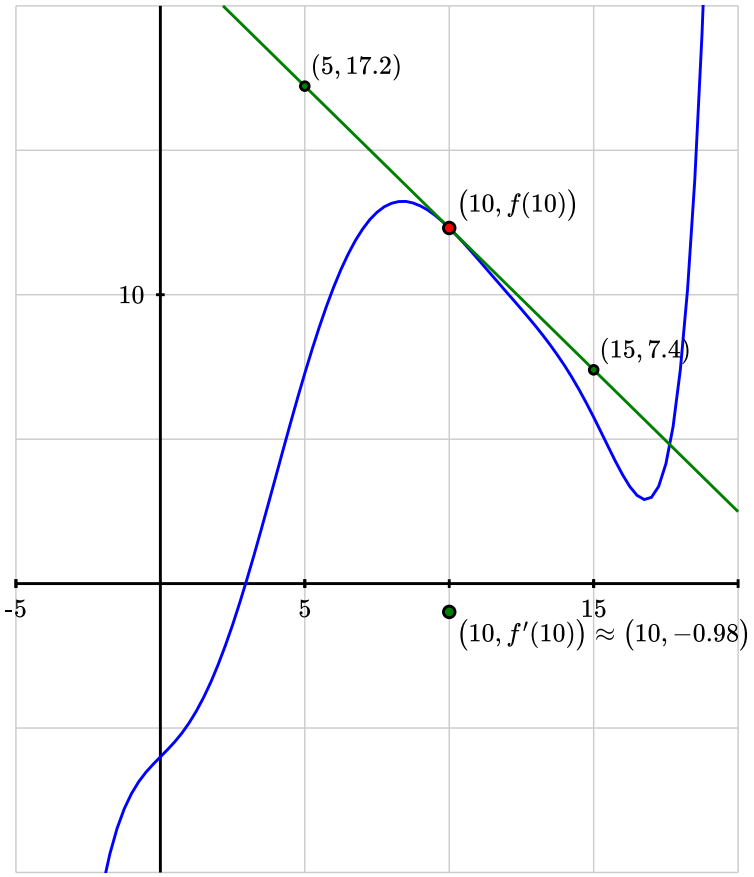


Figure 3.6.1 The graph of $f(x)$ and the tangent line at $a = 10$.

1. Find at least six more points on the graph of the instantaneous rate of change of f by following this procedure:

(a) Pick a point on the graph of f and sketch in a tangent line to the graph at that point.

(b) Find two points on the tangent line that are relatively easy to approximate and use them to estimate the slope of the tangent line.

(c) Plot the point determined by the x-coordinate of the point you chose, and use the slope of the tangent line as the y-coordinate.

2. Now, let's look for connections between the graph of the instantaneous rate of change and the graph of the original function. *Be careful in the following questions to pay attention to whether we are talking about f or its instantaneous rate of change!*

(a) Where is the graph of f increasing most rapidly? What do you notice about the value of the instantaneous rate of change there?

(b) Where is the graph of f (essentially) horizontal? What do you notice about the value of the instantaneous rate of change there?

(c) Over what range(s) of x values do the values of the instantaneous rate of change appear to be increasing, and over what range(s) of x values do the values of the instantaneous rate of change appear to be decreasing? What do you notice about the difference in the shape of the graph of f over those regions?

(d) What other relationships between the graphs of f and the instantaneous rate of change do you notice?

3. Use the information you have found and we have discussed to sketch the rest of the graph of the instantaneous rate of change (that is, connect the dots).

3.7 Derivative Patterns 1: Powers of x

In this activity, we will explore derivatives for functions of the form $f(x) = x^r$ where r is any real number. We will start with some examples and then give exercises designed to help us see a pattern.

Example 3.7.1 We will find the derivative of $f(x) = x^5$ using the definition of the derivative.

We know that for any point x, we can find the derivative (the instantaneous rate of change) of a function $f(x)$ by calculating $f'(x) = \lim_{u\to x} \frac{f(u)-f(x)}{u-x}$. Thus, for our function, we have

$$\begin{aligned} f'(x) &= \lim_{u\to x} \frac{f(u)-f(x)}{u-x} \\ &= \lim_{u\to x} \frac{u^5-x^5}{u-x}. \end{aligned}$$

Using the algebraic identity $u^5 - x^5 = (u-x)\left(u^4+u^3x+u^2x^2+ux^3+x^4\right)$, we have

$$\begin{aligned} f'(x) &= \lim_{u\to x} \frac{u^5-x^5}{u-x}. \\ &= \lim_{u\to x} \frac{(u-x)\left(u^4+u^3x+u^2x^2+ux^3+x^4\right)}{u-x}. \end{aligned}$$

For every value of $u \neq x$, $u - x \neq 0$, which means $\frac{u-x}{u-x} = 1$. Therefore, for every $u \neq x$,

$$\frac{(u-x)\left(u^4+u^3x+u^2x^2+ux^3+x^4\right)}{u-x} = \left(u^4+u^3x+u^2x^2+ux^3+x^4\right).$$

Thus,

$$\begin{aligned} f'(x) &= \lim_{u\to x} \frac{(u-x)\left(u^4+u^3x+u^2x^2+ux^3+x^4\right)}{u-x} \\ &= \lim_{u\to x} \left(u^4+u^3x+u^2x^2+ux^3+x^4\right) \\ &= x^4+x^3x+x^2x^2+xx^3+x^4 \\ &= x^4+x^4+x^4+x^4+x^4 \\ &= 5x^4. \end{aligned}$$

And so, we have found that if $f(x) = x^5$, then $f'(x) = 5x^4$. ▲

Example 3.7.2 We will find the derivative of $f(x) = x^{3/5}$ using the definition of the derivative.

For this function, we have

$$\begin{aligned} f'(x) &= \lim_{u\to x} \frac{f(u)-f(x)}{u-x} \\ &= \lim_{u\to x} \frac{u^{3/5}-x^{3/5}}{u-x}. \end{aligned}$$

We know that $u^{3/5} = \left(u^{1/5}\right)^3$, and so, we know that we can factor $u^{3/5} - x^{3/5}$ as $u^{3/5} - x^{3/5} = \left(\left(u^{1/3}\right) - \left(x^{1/3}\right)\right)\left(\left(u^{1/3}\right)^2 + \left(u^{1/3}\right)\left(x^{1/3}\right) + \left(x^{1/3}\right)^2\right)$, so

$$\begin{aligned} f'(x) &= \lim_{u\to x} \frac{u^{3/5}-x^{3/5}}{u-x} \\ &= \lim_{u\to x} \frac{\left(\left(u^{1/5}\right) - \left(x^{1/5}\right)\right)\left(\left(u^{1/5}\right)^2 + \left(u^{1/5}\right)\left(x^{1/5}\right) + \left(x^{1/5}\right)^2\right)}{u-x} \\ &= \lim_{u\to x} \frac{\left(\left(u^{1/5}\right) - \left(x^{1/5}\right)\right)\left(\left(u^{2/5}\right) + \left(u^{1/5}\right)\left(x^{1/5}\right) + \left(x^{2/5}\right)\right)}{u-x} \end{aligned}$$

Based on previous work, we know that

$$\left(\left(u^{1/5}\right) - \left(x^{1/5}\right)\right)\left(\left(u^{1/5}\right)^4 + \left(u^{1/5}\right)^3\left(x^{1/5}\right) + \left(u^{1/5}\right)^2\left(x^{1/5}\right)^2 + \left(u^{1/5}\right)\left(x^{1/5}\right)^3 + \left(x^{1/5}\right)^4\right) = \left(u^{1/5}\right)^5 - \left(x^{1/5}\right)^5$$

$$= (u - x)\,,$$

so,

$$\begin{aligned}
f'(x) &= \lim_{u\to x} \frac{\left(u^{1/5} - x^{1/5}\right)\left(u^{2/5} + u^{1/5}x^{1/5} + x^{2/5}\right)}{u - x} \\
&= \lim_{u\to x} \frac{(u - x)\left(u^{2/5} + u^{1/5}x^{1/5} + x^{2/5}\right)}{(u - x)\left(u^{4/5} + u^{3/5}x^{1/5} + u^{2/5}x^{2/5} + u^{1/5}x^{3/5} + x^{4/5}\right)} \\
&= \lim_{u\to x} \frac{\left(u^{2/5} + u^{1/5}x^{1/5} + x^{2/5}\right)}{\left(u^{4/5} + u^{3/5}x^{1/5} + u^{2/5}x^{2/5} + u^{1/5}x^{3/5} + x^{4/5}\right)} \\
&= \frac{\left(x^{2/5} + x^{1/5}x^{1/5} + x^{2/5}\right)}{\left(x^{4/5} + x^{3/5}x^{1/5} + x^{2/5}x^{2/5} + x^{1/5}x^{3/5} + x^{4/5}\right)} \\
&= \frac{\left(x^{2/5} + x^{1/5}x^{1/5} + x^{2/5}\right)}{\left(x^{4/5} + x^{3/5}x^{1/5} + x^{2/5}x^{2/5} + x^{1/5}x^{3/5} + x^{4/5}\right)} \\
&= \frac{\left(x^{2/5} + x^{2/5} + x^{2/5}\right)}{\left(x^{4/5} + x^{4/5} + x^{4/5} + x^{4/5} + x^{4/5}\right)} \\
&= \frac{3x^{2/5}}{5x^{4/5}} \\
&= \frac{3}{5}x^{-2/5}
\end{aligned}$$

So, if $f(x) = x^{3/5}$, we have found that $f'(x) = \frac{3}{5}x^{-2/5}$. ▲

1. Let $f(x) = 4$. Use the definition of derivative to find $f'(x)$.

2. Let $f(x) = x$. Use the definition of derivative to find $f'(x)$.

3. Use the definition of the derivative to find $f'(x)$ when $f(x) = x^3$.

4. Let $f(x) = x^{-1} = 1/x$. Use the definition of derivative to find $f'(x)$. Write your answer as $f'(x) = cx^r$.

5. Use the definition of the derivative to find $f'(x)$ when $f(x) = x^{2/3}$. Write your answer as $f'(x) = cx^r$.

6. Let $f(x) = x^{8/3}$. Predict the value of $f'(x)$. Justify your answer.

3.8 What's Natural about the Natural Exponential?

Ever wonder where the "natural" in the natural logarithm and the natural exponential function come from?

In this activity we will explore how the natural exponential function arises in a natural way from derivatives of polynomial functions combined with the idea of an "infinite" polynomial, which is how we are going to think about power series.

3.8.1 A Function that is its Own Derivative

We have seen that sometimes the derivative of a function is *simpler* than the function itself (for example, if $h(x) = 5x^2$, its derivative, $h'(x) = 10x$ is arguably simpler because linear functions are simpler than quadratics), and sometimes the derivative of a function is *more complicated* than the original function (for example, if $g(x) = \sqrt{3x^2+1}$, then the derivative, $g'(x) = \frac{3x}{\sqrt{3x^2+1}}$ is arguably more complicated).

It is reasonable, then, to ask if there are any functions, p, that have the property that $p'(x) = p(x)$.

3.8.1.1 Preliminary Ideas

Before starting, we will review some notation and look at some important patterns in derivatives.

Definition 3.8.1 The Factorial Function. Given any positive integer (that is, a positive whole number), n, the *factorial* of n, written $n!$ and read as "n factorial," is the product of n with all smaller positive integers. That is,

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \cdots \cdot 2 \cdot 1.$$

Here are some examples:

$$\begin{aligned} 3! &= 3 \cdot 2 \cdot 1 \\ &= 6 \end{aligned} \qquad \begin{aligned} 6! &= 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \\ &= 720. \end{aligned}$$

◆

3.8.1.2 An "Infinite" Polynomial

In order to find a our function that is *exactly the same* as its own derivative, we are going make some assumptions.

First of all, a particularly clever calculus student might come up with the zero function as one such function, and it's also pretty easy to see that if $p'(x) = p(x)$, then for any constant c, $(c \cdot p(x))' = c \cdot p'(x) = (c \cdot p(x))$, so we are going to add an extra condition. So, our goal in this activity is to find a function p so that

1. For every real x, $p'(x) = p(x)$, and
2. $p(0) = 1$.

Furthermore, we are going to further assume that our function is represented by a *power series*. We will ignore the issue of convergence for now. Thus, we assume that our function p can be expressed as

$$\begin{aligned} p(x) &= \sum_{k=0}^{\infty} a_k x^k \\ &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots \end{aligned} \tag{3.8.1}$$

for some (constant, but as yet unspecified) real numbers a_0, a_1, a_2, a_3, and so forth.

These assumptions will allow us to find values of a_0, a_1, a_2, a_3, and so forth, and thus completely define the function p.

Let $p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \cdots$.

Exercises

1. Evaluate $p(0)$ for our function p (defined in (3.8.1)). Since we stipulated that $p(0) = 1$, what information do we now have about the values of the coefficients in the power series for p?

2. Calculate $p'(x)$ using (3.8.1), and evaluate $p'(0)$. Because $p'(x) = p(x)$ for all real x, we know that $p'(0) = p(0) = 1$. What information do we now have about the power series for p?

3. Calculate $p''(x)$, and then evaluate at 0. What information do we now have about the power series for p?

4. Continue the process for higher and higher order derivatives. What pattern(s) do you see?

5. Put at least a degree 9 polynomial matching your pattern into an electronic device capable of graphing.

1. Graph your polynomial in the window $-3 \leq x \leq 3, -5 \leq y \leq 20$.
2. Evaluate your polynomial at $x = 1.0$ (in other words, find $p(1.0)$). Does the number look familiar to you?
3. Compare the value of $p(2.0)$ with the value of $(p(1.0))^2$. Compare the value of $p(3.0)$ with the value of $(p(1.0))^3$.
4. Compare $p(2.0)$ with $p(0.5) \cdot p(1.5)$.
5. Graph $y = p(x)$ for $-1 \leq x \leq 2$.

What kind of function does p appear to be?

3.8.1.3 Properties of Exponential Functions

In this part of our exploration, we will show that p must be an *exponential* function.

What exactly does it mean to be an exponential function? In other words, aside from being able to write a function as $f(x) = k^x$, how can we tell if a function is an exponential function?

First, let's start by thinking about exponents more generally.

Checkpoint 3.8.2 Write down as many properties of exponents as you can think of. (For example, $a^c b^c = (ab)^c$.)

Checkpoint 3.8.3 Use properties of exponents to simplify $(3^a)^2$. Specifically, find an expression s so that $(3^a)^2 = 3^s$.

Checkpoint 3.8.4 What is the value of u so that $3^u = 3$?

Checkpoint 3.8.5 What is a number r so that $r^2 = 3$? (Do *not* write this answer in exponential form: that's the next question! Besides, there is a more common notation.)

Checkpoint 3.8.6 Since $\left(\sqrt{3}\right)^2 = 3$, $(3^a)^2 = 3^{2a}$, and $3^1 = 3$, what must be the value of a so that $(3^a)^2 = 3$?

Checkpoint 3.8.7 What is another way to express $3^{3/2}$?

Checkpoint 3.8.8 Does the expression $3^{\ln(2)}$ make sense? How would you define this value?

Notice that if f is an exponential function with base $k > 0$ (that is, for every x, $f(x) = k^x$) we see a couple of important properties. In particular,

- $f(0) = k^0 = 1$,
- $f(1) = k^1 = k$, and
- for all real a and b, $f(a) \cdot f(b) = k^a \cdot k^b = k^{a+b} = f(a + b)$.

Furthermore, we know we can evaluate things like $f(\ln(2))$, but based on our properties of exponents, the way we would do this is with limits, so f must be a continuous function on the reals.

We already know, from the conditions we set when we found it, that our function p satisfies the first condition. We will show that it satisfies the third condition. That is, we will show that $p(a + b) = p(a) \cdot p(b)$. Then we will prove the following theorem.

Theorem 3.8.9 Characterization of Exponential Functions. *For any function f, if*

1. *f is continuous for all real numbers x,*
2. *$f(0) = 1$,*
3. *there is a positive real number r so that $f(1) = r$, and*
4. *for all real a and b, $f(a) \cdot f(b) = f(a + b)$,*

then for any real x, $f(x) = r^x$.

Outline of Proof. We are given that there is a positive real number r so that $f(1) = r$. Define $g(x) = r^x$. Then, in order to prove the theorem, we need to show that for every real x, $f(x) = g(x)$.

- It is clear from our hypotheses that $f(0) = g(0)$ and $f(1) = g(1)$.
- Next, prove that for any positive integer n, $f(n) = g(n)$.
- Then, prove that for any real x, $f(x) \cdot f(-x) = 1$, and thus if $f(x) = g(x)$, then $f(-x) = g(-x)$.
- Prove that for any positive integer n, $f\left(\frac{1}{n}\right) = g\left(\frac{1}{n}\right)$.
- Prove that for any rational number $\frac{p}{q}$, $f\left(\frac{p}{q}\right) = g\left(\frac{p}{q}\right)$.
- Prove that for any real s, $f(s) = g(s)$ using the following two facts:
  1. Continuous functions preserve limits. That is, m is a continuous function if and only if for any sequence a_n, if $\lim_{n\to\infty} a_n = a$ then $\lim_{n\to\infty} m(a_n) = m(a)$.
  2. The rational numbers are *dense* in the reals. That is, for any real number s, there is a sequence of rational numbers s_k so that $\lim_{n\to\infty} s_n = s$.

■

3.8.1.4 A Power Series that Acts Like an Exponential

Based on the ideas developed in subsection Subsubsection 3.8.1.2, p. 34, we define the function p by

$$\begin{aligned} p(x) &= \sum_{k=0}^{\infty} \frac{1}{k!} x^k \\ &= \frac{1}{0!} + \frac{1}{1!}x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \frac{1}{5!}x^5 + \frac{1}{6!}x^6 + \cdots \\ &= 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \frac{1}{120}x^5 + \frac{1}{720}x^6 + \cdots, \end{aligned}$$

and we will show that $p(x)$ satisfies the hypotheses of Characterization of Exponential Functions, and thus is an exponential function.

First, we note that power series are continuous (infinitely differentiable, in fact) at any point for which they are defined. The fact that the power series $p(x)$ is defined for any real number x is a topic for Calculus 2, but we will take it as given, and so $p(x)$ is therefore continuous for all real x.

Next, we have

$$\begin{aligned} p(0) &= \frac{1}{0!} + \frac{1}{1!}0 + \frac{1}{2!}0^2 + \frac{1}{3!}0^3 + \frac{1}{4!}0^4 + \frac{1}{5!}0^5 + \frac{1}{6!}0^6 + \cdots \\ &= 1 + 0 + 0 + 0 + 0 + 0 + 0 + \cdots \\ &= 1. \end{aligned}$$

For the third condition, we need to explain why $p(1)$ is a positive real. It should be clear that

$$p(1) = \sum_{k=0}^{\infty} \frac{1}{k!} 1^k$$

$$\begin{aligned}
&= \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \frac{1}{6!} + \cdots \\
&= 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} + \frac{1}{720} + \cdots \\
&> 0,
\end{aligned}$$

so p satisfies the third condition.

To see that $p(x)$ satisfies the fourth condition, that for all real a and b, $p(a) \cdot p(b) = p(a+b)$, we will evaluate the power series at $a+b$ using the binomial theorem, and show that it is the same as if we multiplied the power series evaluated at a times the power series evaluated at b.

Product of Exponentials Property.

Lemma 3.8.10 The Arithmetic Triangle. *For any non-negative integer n, and any integer k satisfying $0 \leq k \leq n$, the binomial coefficient, defined by $\binom{n}{k} = \frac{n!}{(n-k)!k!}$ satisfies*

- $\binom{n}{0} = 1$;
- $\binom{n}{n} = 1$;
- *for any integer k with $0 < k < n$,*

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof. Let n be any non-negative integer. Then

$$\begin{aligned}
\binom{n}{0} &= \frac{n!}{0!n!} \\
&= \frac{n!}{n!} \\
&= 1
\end{aligned}
\qquad\qquad
\begin{aligned}
\binom{n}{1} &= \frac{n!}{n!0!} \\
&= \frac{n!}{n!} \\
&= 1.
\end{aligned}$$

Now, if k is an integer with $0 < k < n$, then $n > k \geq 1$, so $n \geq 2$, and so $k - 1 \geq 0$ and $n - k \geq 2$, so $(n-1) - k \geq 0$. Thus, $(n-1-k)!$ and $(k-1)!$ are defined. Then

$$\begin{aligned}
\binom{n-1}{k-1} + \binom{n-1}{k} &= \frac{(n-1)!}{((n-1)-(k-1))!(k-1)!} + \frac{(n-1)!}{((n-1)-k)!k!} \\
&= \frac{(n-1)!}{(n-k)!(k-1)!} + \frac{(n-1)!}{(n-k-1)!k!} \\
&= \frac{k \cdot (n-1)!}{(n-k)! \cdot k \cdot (k-1)!} + \frac{(n-k) \cdot (n-1)!}{(n-k)\,(n-k-1)!k!} \\
&= \frac{k \cdot (n-1)!}{(n-k)!k!} + \frac{(n-k) \cdot (n-1)!}{(n-k)!k!} \\
&= \frac{(k+n-k) \cdot (n-1)!}{(n-k)!k!} \\
&= \frac{n \cdot (n-1)!}{(n-k)!k!} \\
&= \frac{n!}{(n-k)!k!} \\
&= \binom{n}{k},
\end{aligned}$$

as was to be shown. ■

Theorem 3.8.11 The Binomial Theorem. *For any elements a, b of a field $\mathbb{F}$, for any non-negative integer n,*

$$\begin{aligned}(a+b)^n &= \sum_{k=0}^{n} \binom{n}{k} a^{n-k}b^k \\ &= \sum_{k=0}^{n} \left(\frac{n!}{(n-k)!k!}\right) a^{n-k}b^k \\ &= \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \left(\frac{n!}{2\cdot(n-2)!}\right)a^{n-2}b^2 + \cdots + \left(\frac{n!}{2\cdot(n-2)!}\right)a^2b^{n-2} + \binom{n}{n-1}ab^{n-1} + \binom{n}{n}b^n \\ &= a^n + na^{n-1}b + \left(\frac{n!}{2\cdot(n-2)!}\right)a^{n-2}b^2 + \cdots + \left(\frac{n!}{2\cdot(n-2)!}\right)a^2b^{n-2} + nab^{n-1} + b^n.\end{aligned}$$

Sketch of Proof. This is a proof by induction where we show that $(a+b)^0 = 1$ and $(a+b)^1 = a+b = 1a+1b$ as our basis steps. We then let $m-1$ be any non-negative integer, assume that our proposition holds for $(a+b)^{m-1}$ and show that because $(a+b)^m = (a+b)(a+b)^{m-1}$, by Lemma 3.8.10, p. 37 and our inductive hypothesis, our proposition holds for $(a+b)^m$. ■

Checkpoint 3.8.12

(a) Start by evaluating the power series $p(x)$ at $a+b$. Expand each line:

$$\begin{aligned} p(a+b) = \quad & 1 && = \quad 1 \\ & + (a+b) && + a + b \\ & + \frac{1}{2}(a+b)^2 && + \frac{1}{2}a^2 + ab + \frac{1}{2}b^2 \\ & + \frac{1}{6}(a+b)^3 && + \underline{\qquad\qquad\qquad\qquad} \\ & + \frac{1}{24}(a+b)^4 && + \underline{\qquad\qquad\qquad\qquad} \\ & + \frac{1}{120}(a+b)^5 && + \underline{\qquad\qquad\qquad\qquad} \\ & + \frac{1}{720}(a+b)^6 && + \underline{\qquad\qquad\qquad\qquad} \\ & + && + \\ & \vdots && \vdots \end{aligned}$$

(b) Next, rearrange the terms in a grid with terms with each power of $(a+b)$ on a diagonal, from lower left to upper right. That is, in the upper left corner we have 1 ($(a+b)^0$), then on the first diagonal, we have $+a$ and $+b$ (from $(a+b)^1$). On the next diagonal, we put the terms from $\frac{1}{2}(a+b)^2$: $\frac{1}{2}a^2$, then ab, then $\frac{1}{2}b^2$.

(c) Continue the pattern up to at least $+\frac{1}{720}(a+b)^6$.

$$\begin{array}{rccccc} p(a+b) = & 1 & +b & +\dfrac{1}{2}b^2 & + & \\ & +a & +ab & + & & \\ & +\dfrac{1}{2}a^2 & + & + & & \\ & + & + & & & \\ & + & & & & \\ & + & & & & \\ & + & & & & \\ & + & & & & \\ & + & & & & \\ & \vdots & & & & \end{array}$$

(d) Now, compare the previous result to our result if we use an array to multiply the series expansion for $p(a)$ by the series expansion for $p(b)$. (Fill in the following and compare it to the above.)

$\cdot$	1	$+b$	$+\frac{1}{2}b^2$	$+\frac{1}{6}b^3$	$+\frac{1}{24}b^4$	$+\frac{1}{120}b^5$	$+\frac{1}{720}b^6$	$+\cdots$
1	1	$+b$	$+\frac{1}{2}b^2$					
$+\,a$	$+a$	$+ab$						
$+\,\frac{1}{2}a^2$	$+\frac{1}{2}a^2$							
$+\,\frac{1}{6}a^3$								
$+\,\frac{1}{24}a^4$								
$+\,\frac{1}{120}a^5$								
$+\,\frac{1}{720}a^6$								
$+$ $\vdots$								

What do the preceding exercises suggest about $p(a)\cdot p(b)$ and $p(a+b)$? What do we now know (from Theorem 3.8.9, p. 36) about $p(x)$?

3.8.2 Synthesis

1. What is something new you learned in this activity, or what is a new insight or way of looking at things that you have developed because of this activity?
2. How do you think Leonard Euler hit upon the idea of the natural exponential function and the natural logarithm?
3. How does the use of power series inform your understanding of the natural exponential function?
4. How could you demonstrate understanding of the material in this activity?
5. How does this activity inform your understanding of the nature of (modern) mathematics?

3.9 Modeling and Optimizing an Origami *Masu* Box

The *masu* is both a traditional unit of measurement (approximately 180 mL) used in Japan for *measuring* sake or rice, and a square-based wooden box traditionally used for *serving* rice or sake.

There is also an origami construction called a *masu* box. We will experiment to see if we can fold a *masu* box so that it would hold as much rice as possible.

First we will fold a "standard" *masu* box together, then, in groups, you will fold several *masu* boxes whose heights (and therefore widths) are different from that of the standard *masu* box and analyze the results.

3.9.1 Procedure

Figure 3.9.1, p. 40 through Figure 3.9.10, p. 40 is a step-by-step pictorial guide to help you fold the boxes. The crucial steps for making boxes of different heights are ***Step 3*** and ***Step 5***. If you fold short of the center the box will be shorter (and wider). If you fold past the center the box will be taller (and narrower).

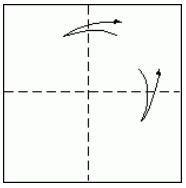


Figure 3.9.1 Step 1: Crease and return twice

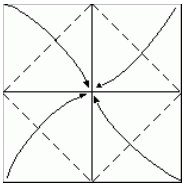


Figure 3.9.2 Step 2: Fold four corners to center

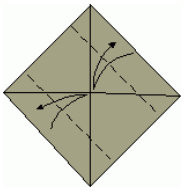


Figure 3.9.3 ***Step 3: Fold two sides to height and return***

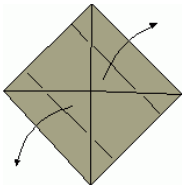


Figure 3.9.4 Step 4: Open two corners

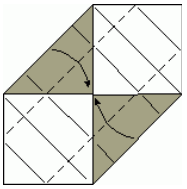


Figure 3.9.5 ***Step 5: Fold other sides to height***

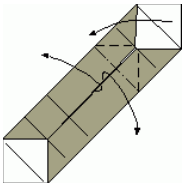


Figure 3.9.6 Step 6: Lift both sides and one end

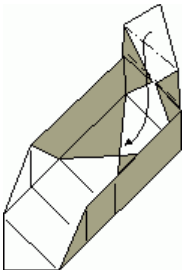


Figure 3.9.7 Step 7: Fold flap to center

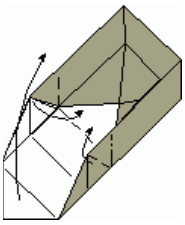


Figure 3.9.8 Step 8: Raise other end

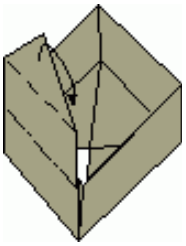


Figure 3.9.9 Step 9: Fold flap to center

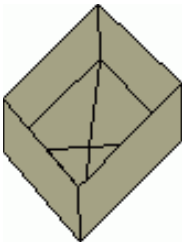


Figure 3.9.10 Step 10: Completed box

Note: The size of the folds in Step 3 and Step 5 must be the same!

- Each group should ***fold at least three boxes*** of differing heights.
- Some suggestions for boxes to fold are:
  - a box of height 1 inch,
  - a box of height 1.5 inches,
  - a box of height 2.5 inches, and
  - a box of height 2.75 inches.

For each of the boxes you fold:

- Plan how to fold the box in order to get the desired height. (For example, you may decide to mark your paper with a pencil to help you choose the positions of your folds.)

- After folding, measure the box carefully to determine its dimensions and calculate its volume. (We will record the data from these measurements as a class.)
- Carefully unfold the box to reveal the fold pattern. (It should look similar to Figure 11, p. 41, 12, p. 41, 13, p. 41.)

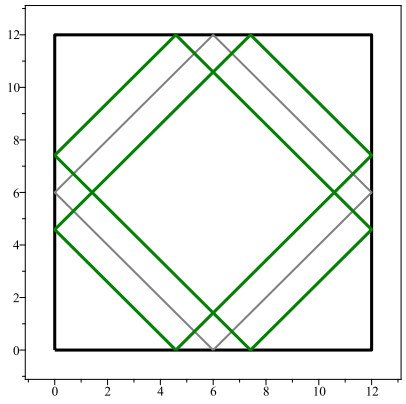


Figure 3.9.11 Short

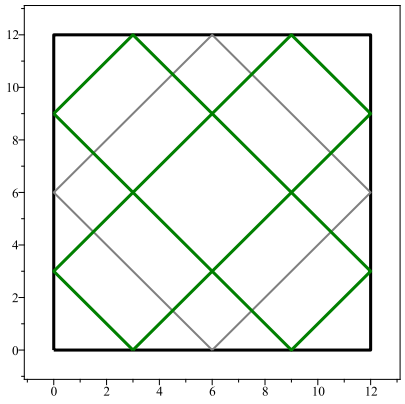


Figure 3.9.12 Standard

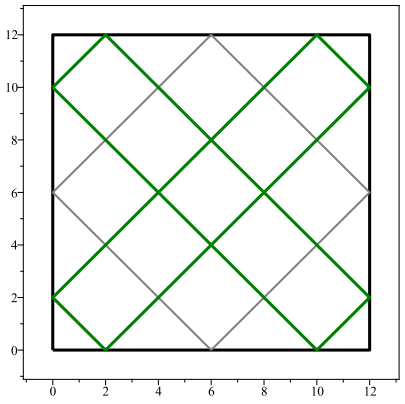


Figure 3.9.13 Tall

3.9.2 Analysis

After examining and measuring the fold patterns for your group's boxes:

1. Make a prediction about the dimensions of an origami *masu* box of maximum volume. (Do you think it will be taller than it is wide, wider than it is tall, or will its height and width be the same? If taller or wider, by how much?)
2. Use the fold pattern to find a relationship between the width and the height (your relationship may involve the length of the diagonal, $d = 12\sqrt{2} \approx 16.97056$ in.)
3. Determine your *decision variable*. ***Stop here to discuss.***
4. Turn your relationship into a function, f, returning the other dimension as a function of the decision variable. That is, your function, f, should satisfy one of $w = f(h)$ or $h = f(w)$.
5. Use your function, f, as a starting point to develop a *volume* function, g, so that the volume, V, is given as a function of the decision variable.
6. Determine what values of the decision variable are reasonable inputs for your functions (f and g) in this situation.
7. Use calculus to find the value of the height that will make the volume as large as possible.
8. Is the "standard" box optimal? If not, is the optimal box taller or shorter?
9. How does your result compare to your prediction? Explain what might account for any discrepancies, or, if your prediction was accurate, explain what insight led to your prediction.
10. If you wish (and you have enough paper) fold a box of the optimal dimensions.

3.10 Identifying Graphs of Derivatives

Objectives

- The learner will use connections between the shape of the graph of a function and its derivative to discern which function could or could not be the derivative of which.

1.

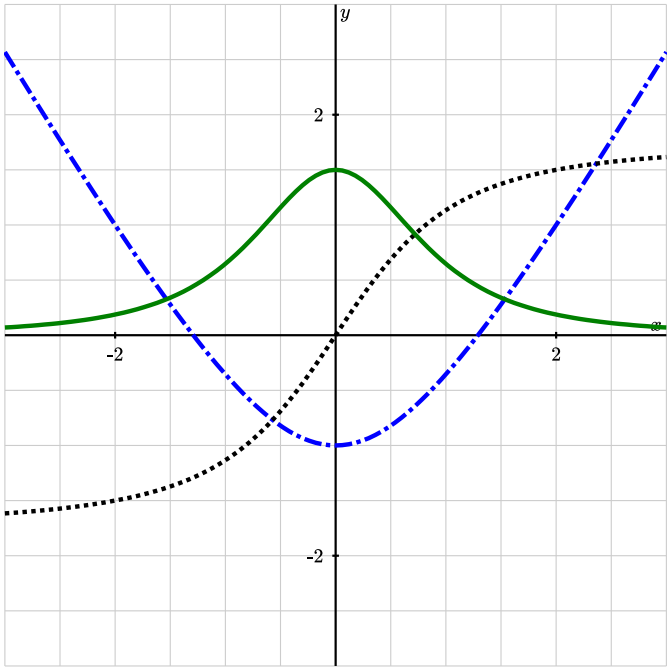


Figure 3.10.1 The graphs $y = \ell(x)$, $y = M(x)$, and $y = k(x)$.

The graphs of three functions are shown in Figure 3.10.1, p. 42. One of these graphs is $y = f(x)$, a second one is the graph of the derivative of $f(x)$, $y = f'(x)$. The third one is the graph of the derivative of $f'(x)$, $y = f''(x)$.

Identify which of the graphs, green solid, blue dashed, or black dotted, is $f(x)$, which is $f'(x)$, and which is $f''(x)$. Explain your choices clearly in terms of function values and rates of change or slopes!

2.

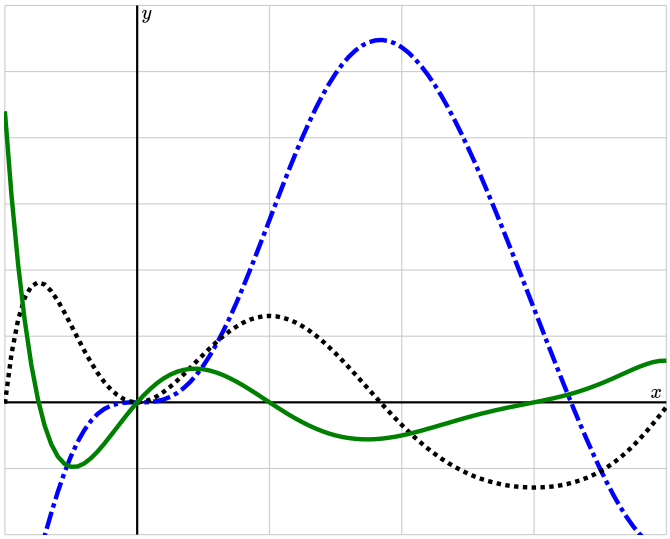


Figure 3.10.2 The graphs $y = h(x)$, $y = T(x)$, and $y = s(x)$.

The graphs of three functions are shown in Figure 3.10.1, p. 42. One of these graphs is $y = g(x)$, a second one is the graph of the derivative of $g(x)$, $y = g'(x)$. The third one is the graph of the derivative of $g'(x)$, $y = g''(x)$.

Identify which of the graphs, green solid, blue dashed, or black dotted, is $g(x)$, which is $g'(x)$, and which is $g''(x)$. Explain your choices clearly in terms of function values and rates of change or slopes!

3.11 Patterns and Properties (Graphs of Derivatives)

Objectives

- The Learner will investigate graphs of derivatives and use their understanding of graphs to develop strategies for finding the derivatives of trigonometric functions.

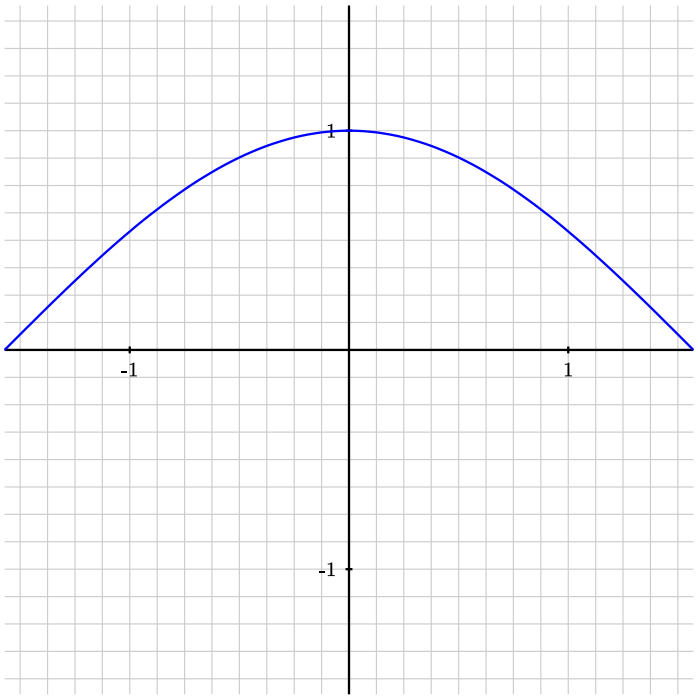


Figure 3.11.1 The graph of $y = f(x)$

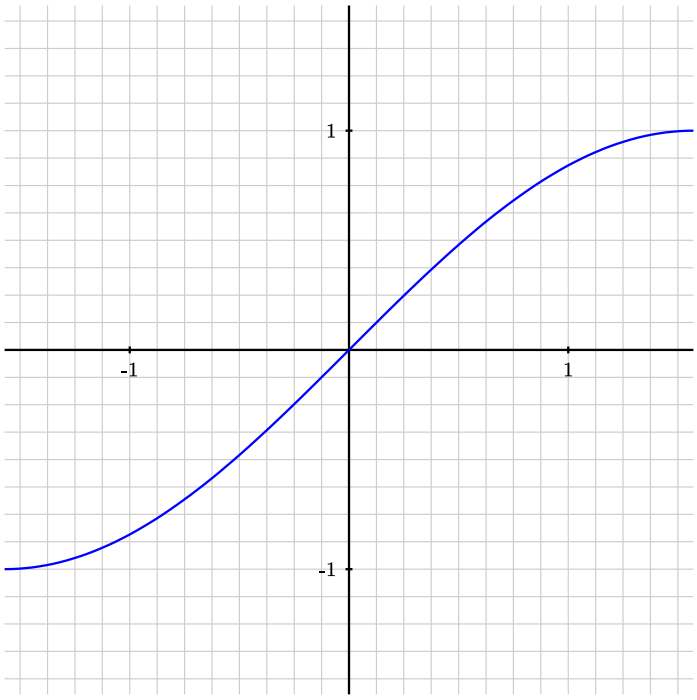


Figure 3.11.2 The graph of $y = g(x)$

1. Find all the values of x for the graph of $y = f(x)$ where f' appears to be zero and mark the points on the x-axis.

2. Find all the values of x for the graph of $y = f(x)$ where f' appears to be one (1) and for each, mark the point $(x, 1)$.

3. Find all the values of x for the graph of $y = f(x)$ where f' appears to be negative one (-1) and for each, mark the point $(x, -1)$.

4. Sketch a graph of $y = f'(x)$ over the graph of $y = f(x)$.

5. Repeat the above steps for $y = g(x)$.

More of the functions is shown in Figure 3.11.3, p. 45, Figure 3.11.4, p. 45. What familiar functions are these, and what appears to be true about their derivatives?

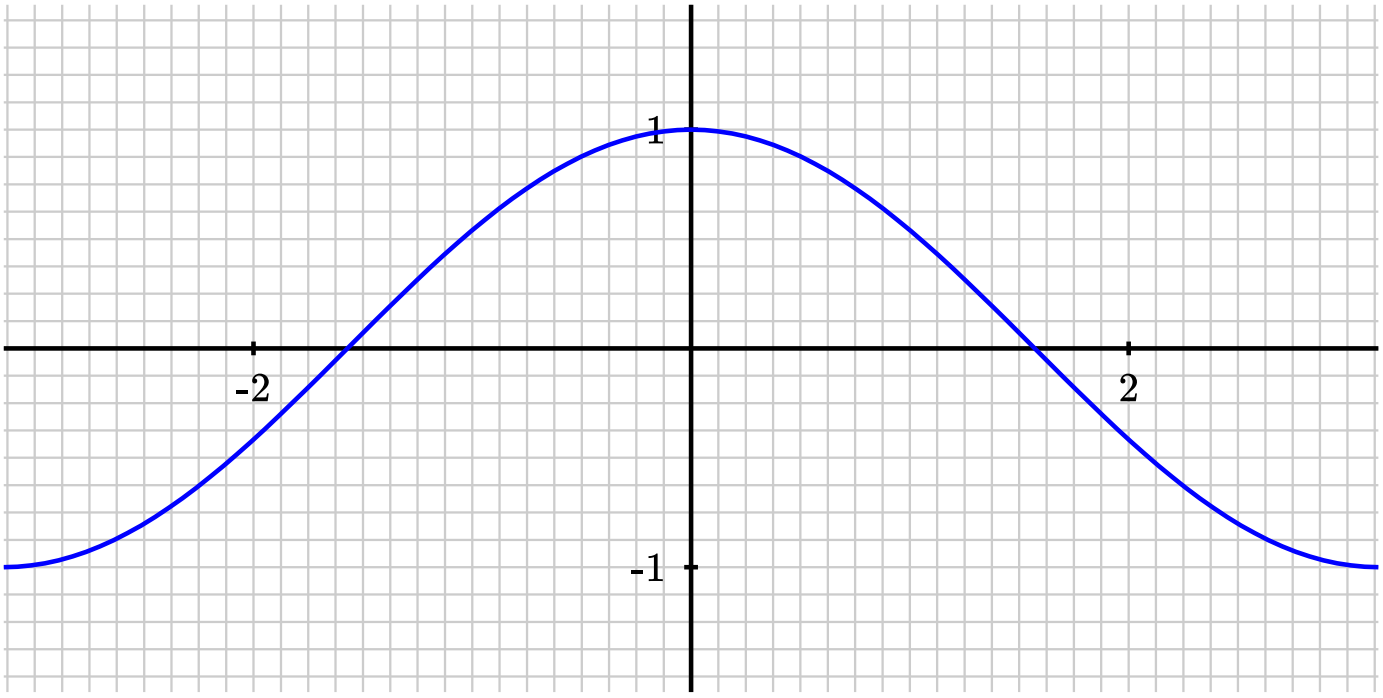


Figure 3.11.3 The graph of $y = f(x)$

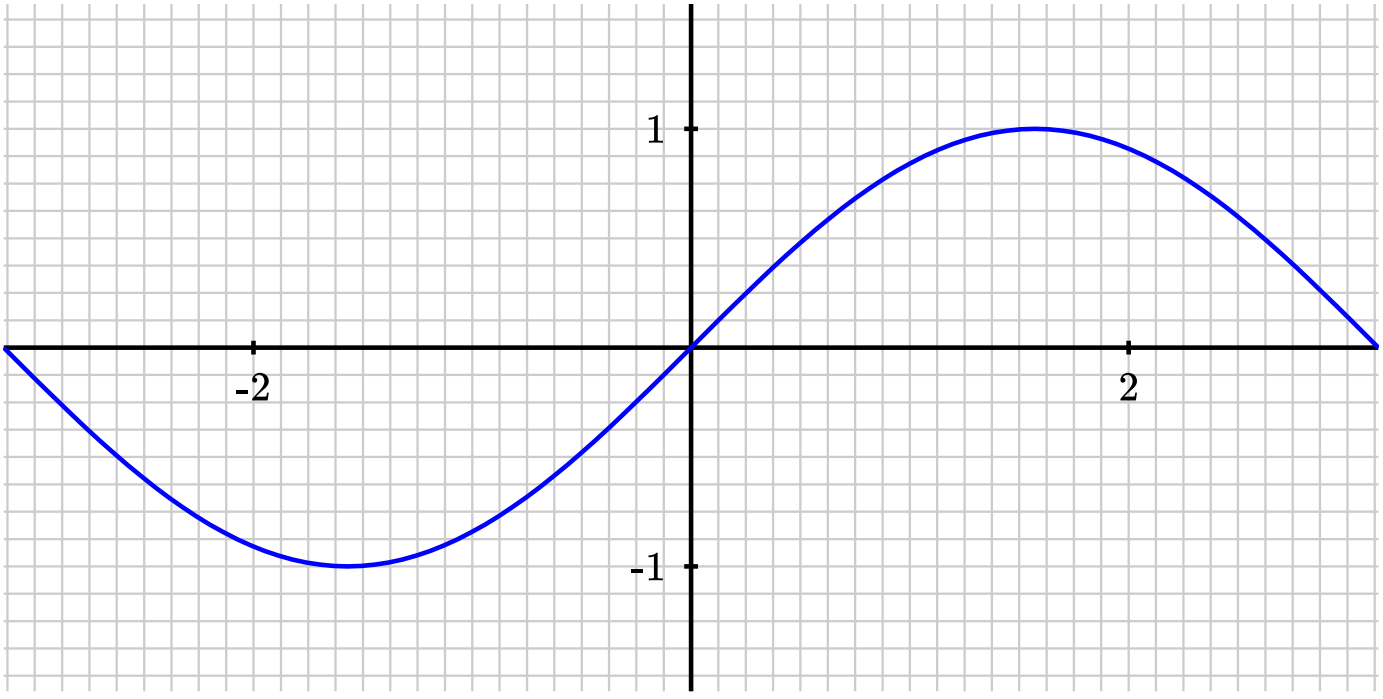


Figure 3.11.4 The graph of $y = g(x)$

3.12 How Many Calculus Students Does it Take to Change a Light Bulb?

Objectives

- This activity is designed to stretch students modeling ability by having them solve a realistic optimization problem.

There is a chandelier in the atrium of the main hall on the campus of Enormous State University that needs to have its light bulbs changed. To get a ladder into the room, maintenance would have to bring it down a hall and around a corner from the maintenance rooms. If the ladder is too long, it will get stuck going around the corner. Figure 3.12.1, p. 46 shows a diagram of the hallway and corner, and examples of various "ladders" that might get stuck.

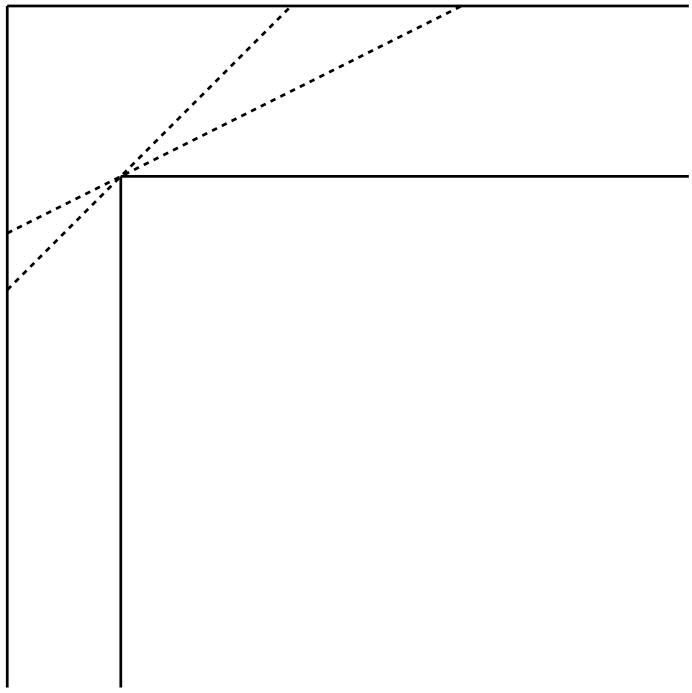


Figure 3.12.1 Two Hallways and Possible "Ladders"

1. The narrower hallway is 2 meters wide, and the wider is 5 meters wide. Maintenance has a really tall employee, Henry, who is tall enough to change the bulbs with their 9 meter ladder (if he stretches), but anyone else would need their next size up, which is 10 meters.

The problem is that a bunch of the bulbs just blew out, and they need the bulbs replaced before the Board of Trustees meets tomorrow. Not knowing that he would be needed, maintenance already allowed Henry to put in his hours for the week, so they would have to pay him overtime.

The alternative, unfortunately, is not very attractive, either. Due to construction, there is no direct way to access the lights without taking a ladder outside, going around a couple of other buildings, and so on, to bring the ladder in the main entrance. This would cost even more than bringing in Henry, because they would have to send *two* employees to carry the ladder that far! (On the other hand, if even the smaller ladder won't fit, they will have to use this option anyway, so they won't add the extra expense of paying Henry overtime!)

Should they give Henry a call?

3.13 Rate of Change for an Area

Objectives

- The learner will investigate the relationship among the derivatives of two functions and the derivative of their product and propose a general method for derivatives of products.

Figure 1, p. 47 shows a model of a computer window being resized. Our measurements started when the window was 13cm wide by 7cm high.

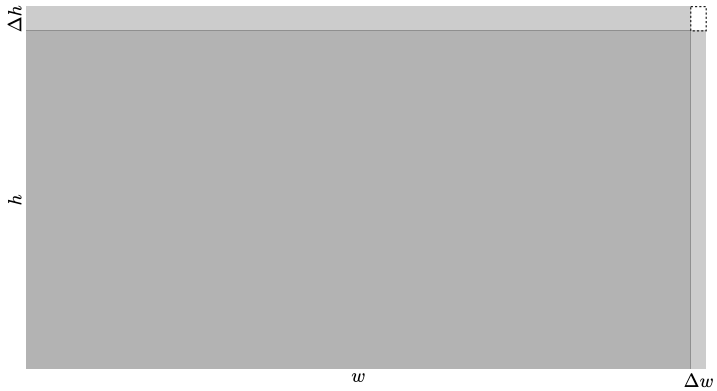


Figure 3.13.1 A Changing Rectangular Region

1. Assuming that *only* the width is changing at a rate of +0.3cm/s, how fast would the total area of the rectangular window be changing?

2. Assuming that *only* the height is changing at a rate of +0.5cm/s, how fast would the total area of the rectangular window be changing?

3. If both changes are happening *simultaneously*, that is the height is increasing at 5cm/s and the width is changing at a rate of +3cm/s, how fast would you expect the total area of the rectangular window be changing? (Justify your answer!)

If we build a model assuming the rates of change in each direction are constant, we might have $A(t) = (13 + 0.3t)(7 + 0.5t) = 13 \cdot 7 + 13 \cdot 0.5t + 0.3t \cdot 7 + 0.3t \cdot 0.5t$. Without simplifying, we can take the derivative (at time $t = 0$ when we start) and get $A'(0) = 13 \cdot 0.5 + 7 \cdot 0.3$. This suggests a pattern for the derivative of a product of functions.

Theorem 3.13.2 The Product Rule. *For two functions, $f(x)$ and $g(x)$, both of which are differentiable at $x = a$, the derivative of their product at $x = a$ is given by*

$$(f \cdot g)' (a) = f'(a)g(a) + f(a)g'(a).$$

Expressed as the derivative function of the product, we have

$$(f \cdot g)' (x) = f'(x)g(x) + f(x)g'(x).$$

Sketch of Proof. We will use the limit definition of derivative for the function $f(x)g(x)$ and add a clever form of zero.

$$\begin{aligned}
(f \cdot g)' (x) &= \lim_{h \to 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\
&= \lim_{h \to 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h} \\
&= \lim_{h \to 0} \frac{f(x+h)g(x+h) - f(x)g(x+h)}{h} + \frac{f(x)g(x+h) - f(x)g(x)}{h} \\
&= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} g(x+h) + f(x) \frac{g(x+h) - g(x)}{h} \\
&= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} g(x+h) + \lim_{h \to 0} f(x) \frac{g(x+h) - g(x)}{h} \\
&= f'(x)g(x) + f(x)g'(x).
\end{aligned}$$

■

3.14 Fuel Efficiency of a Marine Motor

Objectives

- The Learner will investigate how derivatives of functions affect the derivative of new functions obtained by combining them.

The fuel consumption (in gallons per hour) and power output (in horsepower) of some marine (boat) motors are available in their technical specifications as functions of the running speed of the motor (in revolutions per minute).

For the boat motor we are looking at, we will call the \emph{fuel Consumption function} C, and we will call the *Power output fuction* P.

You work at a marine engine company, and your job today is to determine the *Volumetric Fuel Efficiency* (VFE) of an engine and determine how the efficiency changes with engine speed. The Volumetric Fuel Efficiency is the amount of work (in horsepower-hours) that can be done for a specific amount of fuel (so the final units will be horsepower-hours per gallon).

Here is the information you have:

- $C(1800) = 17.61$
- $P(1800) = 61.09$
- $C'(1800) = 0.01776$
- $P'(1800) = 0.07141$

1. Here is an outline of your job for this activity:

(a) Interpret each of the values listed above in the context of this task;

(b) Determine the relationship among C, P, and VFE.

(c) Calculate $VFE(1800)$.

(d) Estimate $VFE'(1800)$ and interpret the results.

3.15 Composition of Functions

In this activity, we review what **composition** of functions means in preparation for investigating patterns for the derivative of a composition of functions.

1. Complete the following tables, given that $h(x) = g\left(f(x)\right)$.

x	$f(x)$
-2	4
-1	
0	
1	5
2	1

Table 3.15.1 Values for $f(x)$

x	$g(x)$
1	
2	1
3	2
4	0
5	-1

Table 3.15.2 Values for $g(x)$

x	$f(x)$
-2	
-1	1
0	2
1	
2	-2

Table 3.15.3 Values for $f(x)$

2. Write each of the following functions as a composition of two or more functions (none of which is allowed to be the **identity function**, $I : x \to x$).

(a) $f : x \to \left(x^3 + 7\right)^5$

(b) $g : x \to \mathrm{e}^{3x-6}$

(c) $k : x \to \frac{1}{\sqrt{x^2+1}}$

(d) $m : x \to \left(x^3 - 3x^2 + 5\right)^{10}$

(e) $\ell : x \to \mathrm{e}^{10x}$

(f) $n : x \to \mathrm{e}^{3x^2}$

3.16 Derivatives of Inverse Functions and Implicit Differentiation

One important application of derivatives of compositions of functions is finding derivatives of functions that are defined only indirectly in terms of other functions. For example, all we know about $\arcsin(x)$ is that it is the inverse of (the principle branch of) the sine. So, how do we find its derivative?

Another instance of this kind of idea is found in progress check 1 where we are given the formula

$$V = 8(h-3)\sqrt{9-(h-3)^2} + 72\arcsin\left(\frac{h-3}{3}\right) + 36\pi$$

for the volume of a water tank (in cubic meters) as a function of the depth of the water (in meters), and we also find the volume as a function of time to be given by $V = 1.1t\text{m}^3$, where t is given in minutes. In this case, we know from our physical experience that the depth of the water is a function of time, but there is no way to directly solve for h as a function of t. Nevertheless, we can use ideas from derivatives of compositions of functions to calculate the rate of change of the depth as a function of time.

3.16.1 Practice with Derivatives of Compositions

Example 3.16.1 The Derivative of a Composition. We will rewrite $f(x) = \sqrt{x^2+4}$ as a composition of functions and find its derivative.

Let $y = g(x) = x^2 + 4$ and let $z = h(y) = \sqrt{y}$. Thus $z = h(y) = h(g(x))$, or equivalently $z = \sqrt{y} = \sqrt{x^2+4}$.

Using Leibniz's notation, we know that $\frac{\mathrm{d}z}{\mathrm{d}x} = \frac{\mathrm{d}z}{\mathrm{d}y}\frac{\mathrm{d}y}{\mathrm{d}x}$. Calculating $\frac{\mathrm{d}z}{\mathrm{d}y}$ and $\frac{\mathrm{d}y}{\mathrm{d}x}$, we have

$$\begin{aligned}
\frac{\mathrm{d}z}{\mathrm{d}x} &= \frac{\mathrm{d}z}{\mathrm{d}y}\frac{\mathrm{d}y}{\mathrm{d}x} \\
&= \frac{\mathrm{d}\sqrt{y}}{\mathrm{d}y}\frac{\mathrm{d}(x^2+4)}{\mathrm{d}x} \\
&= \frac{\mathrm{d}y^{1/2}}{\mathrm{d}y}\frac{\mathrm{d}(x^2+4)}{\mathrm{d}x} \\
&= \left(\frac{1}{2}y^{-1/2}\right)(2x) \\
&= \left(\frac{1}{2\sqrt{y}}\right)(2x) \\
&= \left(\frac{1}{2\sqrt{x^2+4}}\right)(2x) \\
&= \frac{2x}{2\sqrt{x^2+4}} \\
&= \frac{x}{\sqrt{x^2+4}}
\end{aligned}$$

Alternatively, we know that $(h(g(x)))' = h'(g(x))g'(x)$. Now, $h(y) = \sqrt{y} = y^{1/2}$so $h'(y) = \frac{1}{2}y^{-1/2} = 1/(2\sqrt{y})$, and $g(x) = x^2 + 4$, so $g'(x) = 2x$. Therefore,

$$\begin{aligned}
\left(\sqrt{x^2+4}\right)' &= (h(g(x)))' \\
&= h'(g(x))g'(x) \\
&= \frac{1}{2\sqrt{g(x)}}(g'(x)) \\
&= \frac{1}{2\sqrt{x^2+4}}(2x) \\
&= \frac{2x}{2\sqrt{x^2+4}} \\
&= \frac{x}{\sqrt{x^2+4}}.
\end{aligned}$$

▲

Exercises

1. Find the derivative of $\sqrt{x^3+6x}$.

2. Find the derivative of $\sin(x^3+6x)$.

3. Find the derivative of $\sqrt{\sin(x)}$ (assuming $\sin(x)>0$).

4. Find the derivative of e^{x^3+6x}.

5. Let $g(x)$ be an unspecified function. How would you find the derivative of $\sqrt{g(x)}$?

6. Let $g(x)$ be an unspecified function. How would you find the derivative of $g\big(\sin(x)\big)$?

3.16.2 Derivatives of Inverse Functions

Example 3.16.2 We will use composition of functions to find the derivative of $\arcsin(x)$. First of all, we know that $\arcsin(x)$ is the inverse function of the principle branch of the sine. That is, $\sin(\theta)$ with the domain restricted to $-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$. Thus, for any x with $-1\le x\le 1$, if $\theta=\arcsin(x)$, then $\sin(\theta)=x$ and $-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$. In particular, for any x with $-1\le x\le 1$,

$$\sin(\arcsin(x))=x.$$

Let $-1\le x\le 1$, let $y=\arcsin(x)$, and let $z=\sin(y)$. We will now calculate $\frac{\mathrm{d}z}{\mathrm{d}x}$ in two different ways, and then use the fact that the derivative of z must be the same no matter which way we calculate it.

First, we know that $\frac{\mathrm{d}z}{\mathrm{d}x}=\frac{\mathrm{d}z}{\mathrm{d}y}\frac{\mathrm{d}y}{\mathrm{d}x}$. But

$$\begin{aligned}\frac{\mathrm{d}z}{\mathrm{d}y}&=\frac{\mathrm{d}\sin(y)}{\mathrm{d}y}\\&=\cos(y).\end{aligned}$$

Then, we have

$$\begin{aligned}\frac{\mathrm{d}z}{\mathrm{d}x}&=\frac{\mathrm{d}z}{\mathrm{d}y}\frac{\mathrm{d}y}{\mathrm{d}x}\\&=\frac{\mathrm{d}\sin(y)}{\mathrm{d}y}\frac{\mathrm{d}y}{\mathrm{d}x}\\&=\cos(y)\frac{\mathrm{d}y}{\mathrm{d}x}\\&=\cos\big(\arcsin(x)\big)\frac{\mathrm{d}\arcsin(x)}{\mathrm{d}x}.\end{aligned}$$

The value of $\frac{\mathrm{d}\arcsin(x)}{\mathrm{d}x}$ is what we want to find.

Next, we know that $z=\sin\big(\arcsin(x)\big)=x$, so $\frac{\mathrm{d}z}{\mathrm{d}x}=\frac{\mathrm{d}x}{\mathrm{d}x}=1$.

Finally, from trigonometry, we know that if $\sin(\theta)=x$ and $-\pi/2\le\theta\le\pi/2$, then $\cos(\theta)\ge 0$, and so $\cos(\theta)=\sqrt{1-x^2}$. Thus, because $-\pi/2\le\arcsin(x)\le\pi/2$ and $\sin(\arcsin(x))=x$, $\cos(\arcsin(x))=\sqrt{1-x^2}$. Putting all of this together, we have

$$\begin{aligned}\frac{\mathrm{d}z}{\mathrm{d}x}&=\frac{\mathrm{d}z}{\mathrm{d}y}\frac{\mathrm{d}y}{\mathrm{d}x}\\&=\cos\big(\arcsin(x)\big)\frac{\mathrm{d}\arcsin(x)}{\mathrm{d}x}.\\&=\sqrt{1-x^2}\,\frac{\mathrm{d}\arcsin(x)}{\mathrm{d}x}\end{aligned}$$

and

$$\frac{\mathrm{d}z}{\mathrm{d}x}=\frac{\mathrm{d}x}{\mathrm{d}x}$$

$$= 1$$

so $\sqrt{1-x^2}\,\frac{\mathrm{d}\arcsin(x)}{\mathrm{d}x} = 1$, and therefore, for any $-1 < x < 1$,

$$\frac{\mathrm{d}\arcsin(x)}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}}.$$

▲

More generally, we have the following theorem.

Theorem 3.16.3 Derivatives of Inverse Functions. *For any differentiable, invertible function f, with inverse f^{-1}, for any x for which $f'\left(f^{-1}(x)\right) \neq 0$,*

$$\frac{\mathrm{d}f^{-1}(x)}{\mathrm{d}x} = \frac{1}{f'\left(f^{-1}(x)\right)}.$$

Exercises

1. Use the facts that for any $-\pi/2 < x < \pi/2$, $\tan(\arctan(x)) = x$ and $\frac{\mathrm{d}\tan(x)}{\mathrm{d}x} = 1/\cos^2(x)$ to find the derivative of $\arctan(x)$.

2. Use the fact that for any $x > 0$, $\mathrm{e}^{\ln(x)} = x$ to find the derivative of $\ln(x)$.

3. Hyperbolic cosine ($\cosh(x)$) and hyperbolic sine ($\sinh(x)$), which occur frequently in engineering applications, are defined by

$$\cosh(x) = \frac{\mathrm{e}^x + \mathrm{e}^{-x}}{2} \qquad\qquad \sinh(x) = \frac{\mathrm{e}^x - \mathrm{e}^{-x}}{2}$$

(a) Calculate $\sinh(0)$ and $\cosh(0)$.

(b) Show that the derivative of $\sinh(x)$ is $\cosh(x)$.

(c) Show that the derivative of $\cosh(x)$ is $\sinh(x)$.

(d) Verify that for any real number, x:

(i) $\cosh(x) + \sinh(x) = \mathrm{e}^x$.

(ii) $\cosh(x) - \sinh(x) = \mathrm{e}^{-x}$.

(iii) $\cosh^2(x) - \sinh^2(x) = 1$.

(e) Explain why for every real x, $\sinh(x)$ is strictly increasing at x, and therefore has an inverse function, $\operatorname{arcsinh}(x)$.

(f) Show that for any real numbers, x and y, if $\sinh(y) = x$ then $\cosh(y) = \sqrt{1+x^2}$ and therefore $\cosh(\operatorname{arcsinh}(x)) = \sqrt{1+x^2}$.

(g) Let $f(x) = \operatorname{arcsinh}(x)$. Calculate $f'(0)$.

(h) Let $f(x) = \operatorname{arcsinh}(x)$. Calculate $f'(x)$.

3.16.3 Derivative of Implicitly Defined Functions

Example 3.16.4 Let the volume of a water tank (in cubic meters) as a function of depth of the water (in meters) be given by

$$V = 8(h-3)\sqrt{9-(h-3)^2} + 72\arcsin\left(\frac{h-3}{3}\right) + 36\pi$$

and let water be flowing into the (empty) tank at a rate of $1.1\mathrm{m}^3/\mathrm{min}$. We will find the rate at which the water is rising when the water is 2m deep.

Although it is not given explicitly, we know that the depth of the water is a function of time. Furthermore, there is no way to solve for h as a function of time. Thus, we will treat h as an unspecified **implicit function**.

From our work with derivatives of compositions, we know that

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mathrm{d}V}{\mathrm{d}h}\frac{\mathrm{d}h}{\mathrm{d}t}.$$

To calculate $\frac{\mathrm{d}V}{\mathrm{d}h}$, first we will let $k = h - 3$, so that

$$V = 8k\sqrt{9-k^2} + 72\arcsin\left(\frac{k}{3}\right) + 36\pi.$$

Furthermore, $\frac{\mathrm{d}k}{\mathrm{d}h} = \frac{\mathrm{d}(h-3)}{\mathrm{d}h} = 1$, so

$$\begin{aligned}
\frac{\mathrm{d}V}{\mathrm{d}h} &= \frac{\mathrm{d}V}{\mathrm{d}k}\cdot\frac{\mathrm{d}k}{\mathrm{d}h} \\
&= \frac{\mathrm{d}V}{\mathrm{d}k}\cdot 1 \\
&= \frac{\mathrm{d}\left(8k\sqrt{9-k^2}\right)}{\mathrm{d}k} + \frac{\mathrm{d}\left(72\arcsin\left(\frac{k}{3}\right)\right)}{\mathrm{d}k} + \frac{\mathrm{d}36\pi}{\mathrm{d}k} \\
&= \left(8k\frac{\mathrm{d}\left(\sqrt{9-k^2}\right)}{\mathrm{d}k} + \frac{\mathrm{d}\left(8k\right)}{\mathrm{d}k}\sqrt{9-k^2}\right) + 72\left(\frac{1}{\sqrt{1-\left(\frac{k}{3}\right)^2}}\right)\frac{\mathrm{d}\left(\frac{k}{3}\right)}{\mathrm{d}k} + 0 \\
&= \left(8k\frac{1}{2\sqrt{9-k^2}}\frac{\mathrm{d}9-k^2}{\mathrm{d}k} + 8\sqrt{9-k^2}\right) + 72\left(\frac{1}{\sqrt{1-\left(\frac{k}{3}\right)^2}}\right)\frac{1}{3} \\
&= \left(8k\frac{1}{2\sqrt{9-k^2}}(-2k) + 8\sqrt{9-k^2}\right) + 72\left(\frac{3}{\sqrt{9-k^2}}\right)\frac{1}{3} \\
&= 8\left(\frac{-k^2}{\sqrt{9-k^2}} + \sqrt{9-k^2}\right) + \left(\frac{72}{\sqrt{9-k^2}}\right) \\
&= 8\left(\frac{-k^2}{\sqrt{9-k^2}} + \sqrt{9-k^2} + \frac{9}{\sqrt{9-k^2}}\right) \\
&= 8\left(\frac{9-k^2}{\sqrt{9-k^2}} + \sqrt{9-k^2}\right) \\
&= 8\left(\sqrt{9-k^2} + \sqrt{9-k^2}\right) \\
&= 16\sqrt{9-k^2} \\
&= 16\sqrt{9-(h-3)^2}
\end{aligned}$$

Now, we also know that $\frac{\mathrm{d}V}{\mathrm{d}t} = 1.1$, so

$$\begin{aligned}
1.1 &= \frac{\mathrm{d}V}{\mathrm{d}t} \\
&= \frac{\mathrm{d}V}{\mathrm{d}h}\frac{\mathrm{d}h}{\mathrm{d}t} \\
&= 16\sqrt{9-(h-3)^2}\frac{\mathrm{d}h}{\mathrm{d}t}
\end{aligned}$$

and so

$$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{1.1}{16\sqrt{9-(h-3)^2}}.$$

Finally, we evaluate this at $h = 2$:

$$\left.\frac{\mathrm{d}h}{\mathrm{d}t}\right|_{h=2} = \frac{1.1}{16\sqrt{9-(2-3)^2}}$$

$$
\begin{aligned}
&= \frac{1.1}{16\sqrt{9-(-1)^2}} \\
&= \frac{1.1}{16\sqrt{8}} \\
&= \frac{1.1}{32\sqrt{2}} \\
&\approx 0.243067956\text{m/min.}
\end{aligned}
$$

▲

Example 3.16.5 We will find the equation of the tangent line to the ellipse given by $13x^2 + 10xy + 13y^2 = 36$ at the point $\left(0, \frac{6\sqrt{13}}{13}\right)$. (As shown in Figure 3.16.6, p. 55.)

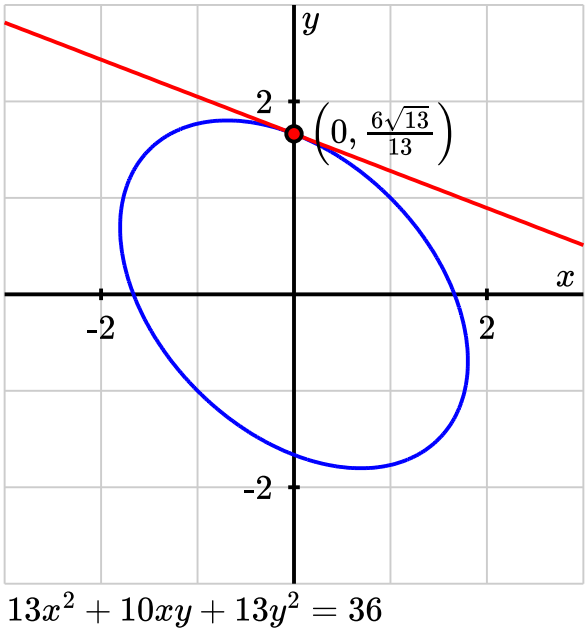


Figure 3.16.6 An Ellipse and a Tangent Line

(a) First, we will verify that the point $\left(0, \frac{6\sqrt{13}}{13}\right)$ is a point on the graph of $13x^2+10xy+13y^2 = 36$ by substituting the x- and y-coordinates into the equation to see if they make the equation true.

$$
\begin{aligned}
13(0)^2 + 10(0)\left(\frac{6\sqrt{13}}{13}\right) + 13\left(\frac{6\sqrt{13}}{13}\right)^2 &= 13\left(\frac{6\sqrt{13}}{13}\right)^2 \\
&= 13\left(\frac{36\cdot 13}{13\cdot 13}\right) \\
&= 36
\end{aligned}
$$

so we see that the point is on the graph.

(b) Next, we will find the **differential** of the left hand side of the equation. We do this by taking the derivative, treating every variable as a function, and ending with the operator d applied to every variable. We will then collect $\mathrm{d}x$ and $\mathrm{d}y$, treating them like variables.

$$
\begin{aligned}
\mathrm{d}\left(13x^2 + 10xy + 13y^2\right) &= 13\,\mathrm{d}\left(x^2\right) + 10\,\mathrm{d}(xy) + 13\,\mathrm{d}\left(y^2\right) \\
&= 13\cdot 2x\,\mathrm{d}x + 10\left(x\,\mathrm{d}y + \mathrm{d}x\,y\right) + 13\cdot 2y\,\mathrm{d}y
\end{aligned}
$$

$$\begin{aligned} &= 26x\,\mathrm{d}x + 10x\,\mathrm{d}y + 10y\,\mathrm{d}x + 26y\,\mathrm{d}y \\ &= (26x+10y)\,\mathrm{d}x + (10x+26y)\,\mathrm{d}y \end{aligned}$$

(c) Now, since the derivative of a constant is zero, $\mathrm{d}(36) = 0$, and so we have $(26x+10y)\,\mathrm{d}x + (10x+26y)\,\mathrm{d}y = 0$. We will now isolate $\frac{\mathrm{d}y}{\mathrm{d}x}$:

$$\begin{aligned} (26x+10y)\,\mathrm{d}x + (10x+26y)\,\mathrm{d}y = 0 &\to (10x+26y)\,\mathrm{d}y = -(26x+10y)\,\mathrm{d}x \\ &\to \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{-(26x+10y)}{(10x+26y)} \end{aligned}$$

(d) We now calculate $\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{(x,y)=(0,6\sqrt{13}/13)}$ to find the slope of the tangent line.

$$\begin{aligned} \left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{(x,y)=(0,6\sqrt{13}/13)} &= \left.\frac{-(26x+10y)}{(10x+26y)}\right|_{(x,y)=(0,6\sqrt{13}/13)} \\ &= \frac{-(26(0)+10(6\sqrt{13}/13))}{(10(0)+26(6\sqrt{13}/13))} \\ &= \frac{-10(6\sqrt{13}/13)}{26(6\sqrt{13}/13)} \\ &= \frac{-5}{13} \end{aligned}$$

(e) Finally, we will use the point-slope form of a line to find the equation of the tangent line: $y = \frac{-5}{13}(x-0) + \frac{6\sqrt{13}}{13}$.

▲

Checkpoint 3.16.7 A train is travelling due east at 90m/s (approximately 201.324mph). An observer is positioned 100m south of a point, P, on the track and is following the train as it approaches from the west with a camera mounted on a tripod.

We want to know the rate at which the camera is turning (in radians per second) when the train is 50m west of P.

(a) Let x be the displacement of the train relative to P (with west being negative and east being positive) and let θ be the angle that the line-of-sight of the camera makes with the line from the observer to P. Find an equation relating x and θ.

(b) Both θ and x can be thought of as functions of time, $\theta(t)$ and $x(t)$. Replace θ with $\theta(t)$ and x with $x(t)$ in your equation. Each side of your equation is also a function of t. Call them $L(t)$ (for left) and $R(t)$ (for right). Take the derivatives of $L(t)$ and $R(t)$ with $\theta(t)$ and $x(t)$ treated as unspecified functions. (Your derivatives will contain $\theta'(t)$ and $x'(t)$.)

(c) Because $L(t) = R(t)$, we must have $L'(t) = R'(t)$. Set $L(t) = R(t)$ and solve for $\theta'(t)$.

(d) Use your understanding of trigonometry to replace any trigonometric functions of $\theta(t)$ with expressions involving $x(t)$.

(e) At the time of interest, $x(t) = -50$m (since the train is 50m west of P), and $x'(t) = +90$m/s (since the train is moving toward the east). Substitute these values into $L'(t) = R'(t)$ and determine the value for $\theta'(t)$.

Checkpoint 3.16.8 Figure 3.16.9, p. 57 shows a curve called an **elliptic curve** given by the equation $y^2 = x^3 - 2x - 1/4$ and a tangent line to the curve.

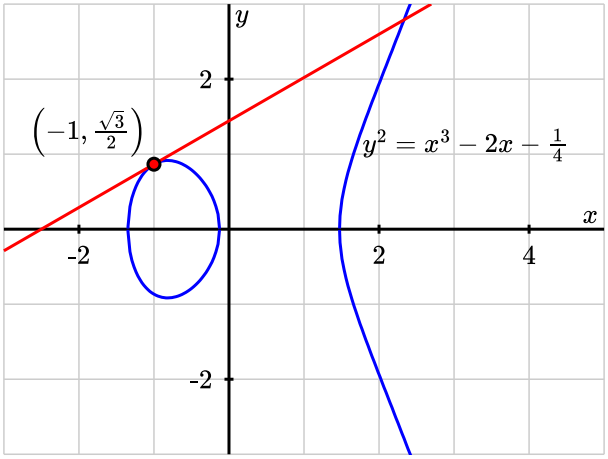


Figure 3.16.9 An Elliptic Curve with a Tangent Line

(a) Verify that the point $(-1, \sqrt{3}/2)$ is on the elliptic curve.

(b) For points near $(-1, \sqrt{3}/2)$, y is an implicit function of x.

3.17 Curve Sketching with Calculus

The learner will use information about a function and its derivative to determine the overall shape of the function's graph.

1. On the axes provided, sketch the graph of a *single function*, f, that has *all* of the following characteristics, briefly explaining what each of the characteristic tells you about the graph.

(a) f is continuous;

(b) $f(-5) = 0$;

(c) $f(0) = 0$;

(d) $f(2) = 4$;

(e) $f(5) = 0$;

(f) $f'(x) < 0$ for $x < -3$;

(g) $f'(-3) = 0$;

(h) $f'(x) > 0$ for $-3 < x < 2$;

(i) $\lim\limits_{x \to 2} \dfrac{f(x) - f(2)}{x - 2}$ does not exist.

(j) $f'(x) < 0$ for $2 < x < 5$;

(k) $f'(5) = 0$;

(l) $f'(x) > 0$ for $5 < x < 7$;

(m) $f'(7)$ is undefined;

(n) $f'(x) > 0$ for $7 < x < 9$;

(o) $f'(9) = 0$;

(p) $f'(x) < 0$ for $x > 9$;

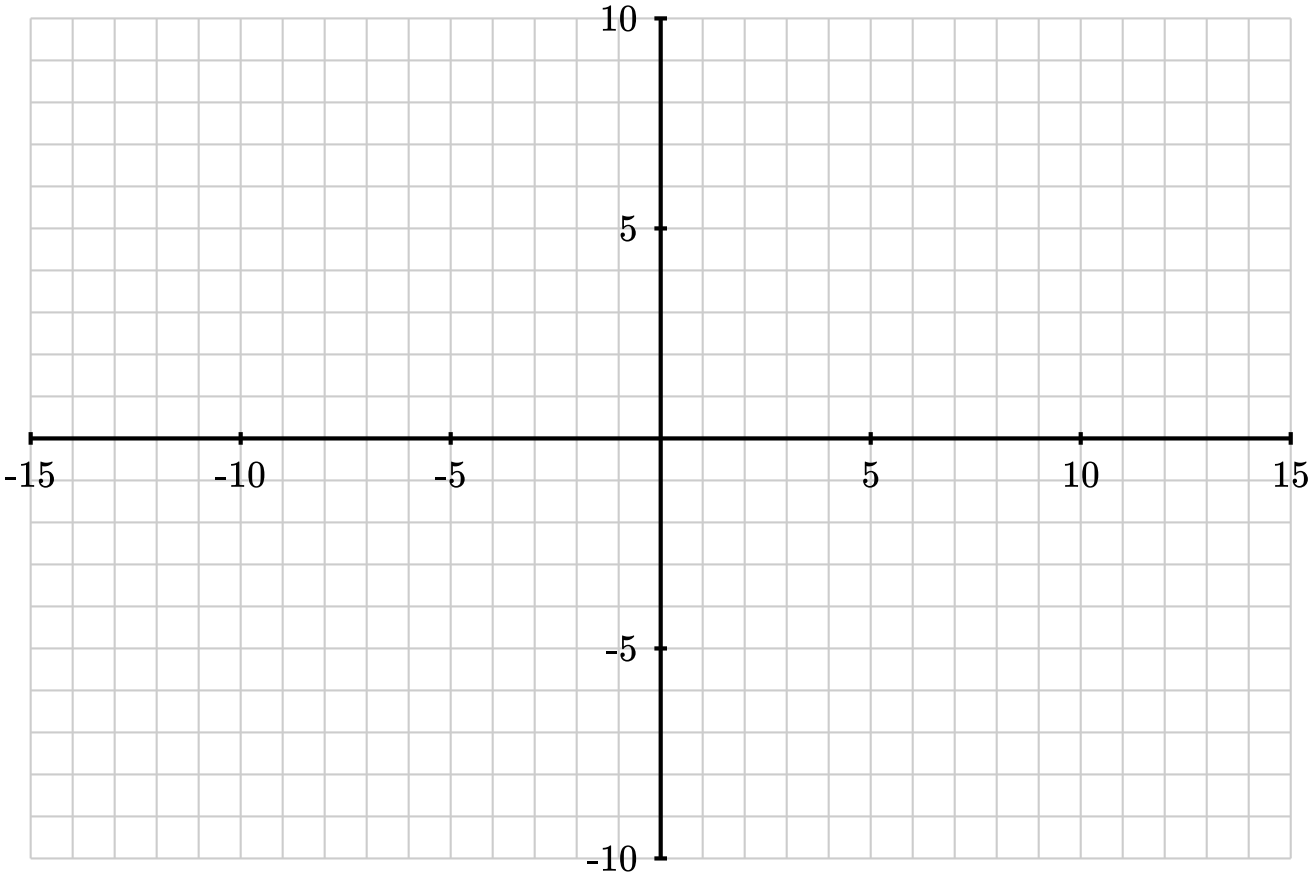
(q) $f'(x)$ decreasing for $x < -4$;

(r) $f''(x) > 0$ for $-4 < x < 2$;

(s) $f'(x)$ increasing for $2 < x < 7$;

(t) $f''(x) < 0$ for $7 < x < 10$;

(u) $f''(x) > 0$ for $x > 10$;


10
5
-15
-10
-5
5
10
15
-5
-10

3.18 Get The Ball, Elvis!

Tim Pennings, a mathematics professor, had a dog named Elvis (a Welsh Corgi), and Tim claims that Elvis seemed to know calculus. According to Tim, Elvis could solve optimization problems — in particular, he could minimize the time it takes to fetch a ball that Tim throws into Lake Michigan for him.

In this activity, we will model and solve the optimization problem using calculus, and then analyze the problem to see if we can understand exactly how Elvis did it.

We will start with the following assumptions:

- Elvis' running speed is 6.40 m/s, and
- Elvis' swimming speed is 1.90 m/s.

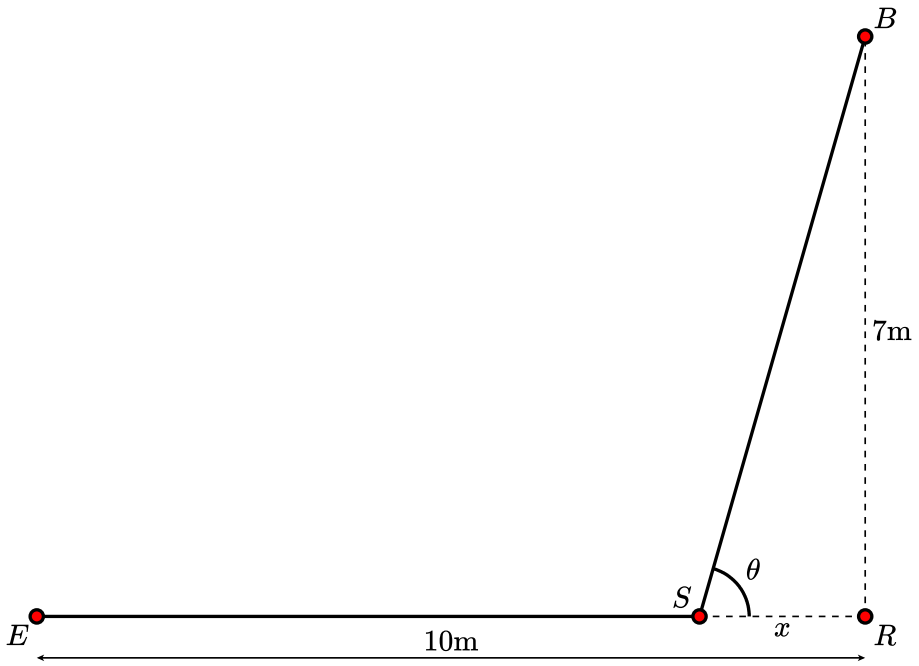


Figure 3.18.1 A possible path for Elvis to follow.

The diagram in Figure 1, p. 60 represents the basic set-up of the optimization problem. Elvis starts at the point marked "E" The ball lands at the point marked "B" The closest point on the shore to where the ball lands is marked "R" (because we will use it as a reference point). After the ball is thrown, Elvis runs some distance (possibly zero meters) along the shore toward the reference point, and then swims directly toward the ball. (That is, runs some distance (possibly zero meters) along the shore toward the reference point, and then swims directly toward the ball. (That is, Elvis *chooses* the point labeled "S".)

Also, so that we are all using the same references, we will define x to be the distance away from R at which Elvis starts swimming. That is, x is the distance between R and S.

We are also interested in the angle at which Elvis swims toward the ball, labeled as θ in the diagram.

1. Beginning with specific distances, find an optimal path, angle, and time, for Elvis to reach the ball. Then, experiment by varying the distances, solve the new problems, and generalize your results.

That is, eventually you want a complete generalization: Assuming that the ball lands at so that the nearest point on the shore is L meters away from Elvis and the ball is w meters away from the shore (where both L and w are *constants*) predict the angle at which Elvis will swim toward the ball and verify your results.

Here is one way you might approach the problem:

(a) Assume that the ball lands 7 meters out from a point 10 meters along the shore.

Develop a formula that gives the time for Elvis to reach the ball as a function of the distance x from the reference point at which Elvis begins to swim.

Assign this function to the variable t .

(b) Graph $y = t(x)$. According to your graph, at approximately what distance, x_* , from the reference point should Elvis start swimming to minimize the time require to reach the ball?

Use calculus terms to describe the feature(s) of the graph that allow you to determine that the point you chose is in fact the minimum. How much time is required to reach the ball? Use calculus terms to describe the feature(s) of the graph that allow you to determine that the point you chose is in fact the minimum. In particular,

(i) what can you say about the value of t' at this point?

(ii) What can you say about the value of t'' at this point?

(c) Calculate the derivative of your time function, then use your answer to above to set up an equation for Maple to your answer to above to set up an equation for Maple to solve. Verify that your solution agrees with your answer to .

(d) Find the time required for Elvis to swim to the ball if he follows the path you described.

(e) Find the angle (θ) at which Elvis will have to swim to follow the path you determined. (Hint: It may be easier to see what is going on if you think about how the *cosine* of the angle is defined.)

2. Assume that the ball lands 7 meters out from a point 15 meters along the shore.

(a) Modify your time function to fit the new situation,

(b) Determine where Elvis should enter the water to reach the ball most quickly, the ball most quickly,

(c) Calculate the time it will take Elvis to reach the ball if he enters the water where you say he should, and

(d) Determine the angle at which Elvis will swim toward the ball.

3. Assume that the ball lands 7 meters out from a point 30 meters along the shore.

(a) Modify your time function to fit the new situation,

(b) Determine where Elvis should enter the water to reach the ball most quickly, the ball most quickly,

(c) Calculate the time it will take Elvis to reach the ball if he enters the water where you say he should, and

(d) Determine the angle at which Elvis will swim toward the ball.

4. Choose several more values for the starting distance between Tim and Elvis. For each, calculate the values of the optimal time, the distance x (the distance away from Tim at which Elvis will start swimming), and the angle, θ at which Elvis will swim. Be systematic, and experiment until you find a pattern.

5. Assume that the ball lands 7 meters out from a point L meters along the shore. (Here we are treating L as a *parameter* . That is, L is constant, but our results may depend on its value.) Based on the pattern you found:

(a) predict the angle at which Elvis will swim toward the ball;

(b) check your prediction with the values of L that you already used. If necessary, modify your model and prediction so that it will work for every possible value of L . (Your model and prediction may have to be *conditional* ; that is, you may have to say things like "if L is less than … ," or "if the angle between Elvis and the ball is … .").

6. Assume that the ball lands w meters out from a point L meters along the shore.

(a) Modify your time function to fit the new situation,

(b) Determine where Elvis should enter the water to reach the ball most quickly, the ball most quickly,

(c) Calculate the time it will take Elvis to reach the ball if he enters the water where you say he should, and

(d) Determine the angle at which Elvis will swim toward the ball.

7. **(Extension)** Re-solve the problem with Elvis' running speed, r, and swimming speed, s, as parameters.

8. **(Extension)** Have Elvis start some distance I away from the shore, create a model, and solve. Verify that the previous situation is a limiting situation for when I approaches zero.

3.19 More Ladders in Corners

Henry has to break in the new person in maintenance this week, and one of the things Henry has had to deal with is taking ladders around corners (again!).

The halls are wide enough that fitting around the corner isn't the problem. The trouble is that the new person doesn't slow down to go around corners! (At least the new person remembers to swing wide and hug the outside wall!)

- The new person walks at a steady 1.2m/s.
- Henry has the new person leading the way with their 10m ladder.
- Assume that the ends of the ladder are always touching the two outer walls of the corner.

1. How fast does Henry have to go to keep up with the new person when Henry is 8m away from the corner?

2. How fast does Henry have to go to keep up with the new person when Henry is 6m away from the corner?

3. What trend do you see in the value of $\mathrm{d}y\,/\,\mathrm{d}t$ as y decreases? Does this make sense?

4. Generalize by finding an expression that gives Henry's speed when Henry is at any distance, d, from the wall.

5. Generalize further by making the other person's speed and the ladder length **parameters**. (That is, they can be any value but are treated as constants within the problem.)

3.20 More Implicit Curves

In the equation $x^3 - y^3 = 18xy$, y is an implicitly function of x (except at $(0,0)$). The graph of $x^3 - y^3 = 18xy$ is shown in Figure 3.20.1, p. 64.

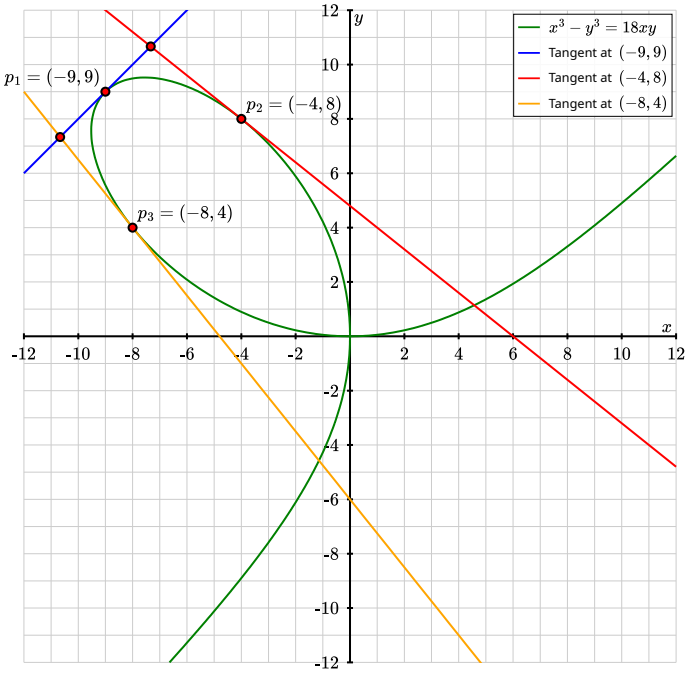


Figure 3.20.1 An Implicit Curve

1. Verify that $(-9,9)$, $(-4,8)$, and $(-8,4)$ are on the curve.
2. Calculate the derivative of y implicitly. That is, find $\frac{dy}{dx}$ for the curve.
3. Find the equations of the tangent lines to the curve for each of the three given points.
4. Find the points where the tangent line at $(-9,9)$ intersects each of the other two tangent lines.
5. Find the coordinates of any points where there is a horizontal tangent line.
6. Find the coordinates of any points where there is a vertical tangent line.
7. For points on the curve near the point $(-9,9)$, estimate the value of y for $x = -8.9$.

Chapter 4

Handouts

4.1 An Introduction to Problem Solving

In his book, *How To Solve It*, mathematician George Polya formulated four steps that apply to any type of problem solving. These steps are *Understand the Problem*, *Make a Plan*, *Carry Out the Plan*, and *Look Back*.

Sometimes it feels like math classes give almost all their attention to carrying out a plan with little attention paid to the other three steps. Often, instructors *give* students the plan rather than asking them to work on *understanding* and *formulating* a plan for themselves, and almost no time is given to *looking back*.

This handout gives more information on each of the four steps, and in future problems we will attempt to apply all four.

Understanding the Problem. ***First.*** you have to *understand* the problem.

What is the unknown? What are the data? What is the condition? Is it possible to satisfy the condition? Is the condition sufficient to determine the unknown? Or is it insufficient? Or redundant? Or contradictory?

Draw a figure. Introduce suitable notation. Separate the various parts of the condition. Can you write them down?

Devising a Plan. ***Second.*** Find the connection between the data and the unknown.

You may be obliged to consider auxiliary problems if an immediate connection cannot be found. You should obtain eventually a *plan* of the solution.

Have you seen it before? Or have you seen the same problem in a slightly different form?

Do you know a related problem? Do you know a theorem that could be useful? *Look at the unknown!* And try to think of a familiar problem having the same or a similar unknown.

Here is a problem related to yours and solved before. Could you use it? Could you use its result? Could you use its method? Should you introduce some auxiliary element in order to make its use possible?

Could you restate the problem? Could you restate it still differently? Go back to definitions.

If you cannot solve the proposed problem try to solve first some related problem. Could you imagine a more accessible related problem? A more general problem? A more special problem? An analogous problem? Could you solve a part of the problem? Keep only a part of the condition, drop the other part; how far is the unknown then determined, how can it vary? Could you derive something useful from the data? Could you think of other data appropriate to determine the unknown? Could you change the unknown or the data, or both if necessary, so that the new unknown and the new data are nearer to each other?

Did you use all the data? Did you use the whole condition? Have you taken into account all essential notions involved in the problem?

Carrying Out the Plan. ***Third.*** *Carry out* your plan.

Carrying out your plan of the solution, *check each step.* Can you see clearly that the step is correct? Can you prove that it is correct?

Looking Back. ***Fourth.*** *Examine* the solution obtained.

Can you *check the result?* Can you check the argument?

Can you derive the result differently? Can you see it at a glance?

Can you use the result, or the method, for some other problem?

4.2 Simple Patterns in Derivatives

We will look at some simple functions, and patters that occur in their derivatives.

Example 4.2.1 Let $f(x) = 4$. That is, no matter what value of $x$we input into $f(x)$, the output will be 4. Then the derivative of $f(x)$ is given by

$$\begin{aligned} f'(x) &= \lim_{u\to x} \frac{f(u)-f(x)}{u-x} \\ &= \lim_{u\to x} \frac{4-4}{u-x} \\ &= \lim_{u\to x} \frac{0}{u-x}. \end{aligned}$$

For every value of $u \neq x$, $\frac{0}{u-x} = 0$.[1] This means that the expression $\frac{0}{u-x}$ is identical with 0 except when $u = x$. Thus,

$$\begin{aligned} f'(x) &= \lim_{u\to x} \frac{f(u)-f(x)}{u-x} \\ &= \lim_{u\to x} \frac{0}{u-x} \\ &= \lim_{u\to x} 0 \\ &= 0. \end{aligned}$$

▲

There is nothing particularly special about 4 in this example. The same result holds for any function whose output is constant. This kind of universal pattern (it holds for all constant functions) is a **theorem**.

Theorem 4.2.2 Derivatives of Constant Functions. *For any constant value c, if $f(x) = c$ (for all values of x), then $f'(x) = 0$.*

Proof. The proof is identical to Example 4.2.1, p. 67 with 4 replaced by c. ■

Theorem 4.2.3 Derivatives of Powers of x. *For any non-zero real number r, if $f(x) = x^r$, then $f'(x) = rx^{r-1}$.*

Proof. We will examine three cases: when r is a positive rational number, when r is a negative rational number, and when r is irrational.

Positive Rational r. Let $f(x) = x^r$ where r is positive and rational, then there are positive integers, p and q so that $r = \frac{p}{q}$, and thus $f(x) = x^{\frac{p}{q}}$. Then

$$\begin{aligned} f'(x) &= \lim_{u\to x} \frac{f(u)-f(x)}{u-x} \\ &= \lim_{u\to x} \frac{u^r - x^r}{u-x} \\ &= \lim_{u\to x} \frac{u^{p/q} - x^{p/q}}{u-x} \\ &= \lim_{u\to x} \frac{\left(u^{p/q}-x^{p/q}\right)\left(\left(u^{p/q}\right)^{q-1} + \left(u^{p/q}\right)^{q-2}\left(x^{p/q}\right) + \left(u^{p/q}\right)^{q-3}\left(x^{p/q}\right)^2 + \cdots + \left(u^{p/q}\right)\left(x^{p/q}\right)^{q-2} + \left(x^{p/q}\right)^{q-1}\right)}{(u-x)\left(\left(u^{p/q}\right)^{q-1} + \left(u^{p/q}\right)^{q-2}\left(x^{p/q}\right) + \left(u^{p/q}\right)^{q-3}\left(x^{p/q}\right)^2 + \cdots + \left(u^{p/q}\right)\left(x^{p/q}\right)^{q-2} + \left(x^{p/q}\right)^{q-1}\right)} \\ &= \lim_{u\to x} \frac{\left(u^{p/q}\right)^q - \left(x^{p/q}\right)^q}{(u-x)\left(\left(u^{p/q}\right)^{q-1} + \left(u^{p/q}\right)^{q-2}\left(x^{p/q}\right) + \left(u^{p/q}\right)^{q-3}\left(x^{p/q}\right)^2 + \cdots + \left(u^{p/q}\right)\left(x^{p/q}\right)^{q-2} + \left(x^{p/q}\right)^{q-1}\right)} \\ &= \lim_{u\to x} \frac{u^p - x^p}{(u-x)\left(\left(u^{p/q}\right)^{q-1} + \left(u^{p/q}\right)^{q-2}\left(x^{p/q}\right) + \left(u^{p/q}\right)^{q-3}\left(x^{p/q}\right)^2 + \cdots + \left(u^{p/q}\right)\left(x^{p/q}\right)^{q-2} + \left(x^{p/q}\right)^{q-1}\right)} \\ &= \lim_{u\to x} \frac{(u-x)\left(u^{p-1} + u^{p-2}x + u^{p-3}x^2 + \cdots + x^{p-1}\right)}{(u-x)\left(\left(u^{p/q}\right)^{q-1} + \left(u^{p/q}\right)^{q-2}\left(x^{p/q}\right) + \left(u^{p/q}\right)^{q-3}\left(x^{p/q}\right)^2 + \cdots + \left(u^{p/q}\right)\left(x^{p/q}\right)^{q-2} + \left(x^{p/q}\right)^{q-1}\right)} \end{aligned}$$

[1] One way to think about this is that in the limit $\lim_{u\to x} \frac{0}{u-x}$, the numerator *is* 0, while the denominator is *approaching* 0.

$$
\begin{aligned}
&= \lim_{u\to x} \frac{\left(u^{p-1}+u^{p-2}x+u^{p-3}x^2+\cdots+x^{p-1}\right)}{\left(\left(u^{p/q}\right)^{q-1}+\left(u^{p/q}\right)^{q-2}\left(x^{p/q}\right)+\left(u^{p/q}\right)^{q-3}\left(x^{p/q}\right)^2+\cdots+\left(u^{p/q}\right)\left(x^{p/q}\right)^{q-2}+\left(x^{p/q}\right)^{q-1}\right)} \\
&= \frac{\left(x^{p-1}+x^{p-2}x+x^{p-3}x^2+\cdots+x^{p-1}\right)}{\left(\left(x^{p/q}\right)^{q-1}+\left(x^{p/q}\right)^{q-2}\left(x^{p/q}\right)+\left(x^{p/q}\right)^{q-3}\left(x^{p/q}\right)^2+\cdots+\left(x^{p/q}\right)\left(x^{p/q}\right)^{q-2}+\left(x^{p/q}\right)^{q-1}\right)} \\
&= \frac{\overbrace{x^{p-1}+x^{p-1}+x^{p-1}+\cdots+x^{p-1}}^{p \text{ copies}}}{\underbrace{\left(\left(x^{p/q}\right)^{q-1}+\left(x^{p/q}\right)^{q-1}+\left(x^{p/q}\right)^{q-1}+\cdots+\left(x^{p/q}\right)^{q-1}+\left(x^{p/q}\right)^{q-1}\right)}_{q \text{ copies}}} \\
&= \frac{px^{p-1}}{q\left(x^{p/q}\right)^{q-1}} \\
&= \frac{p}{q}\frac{x^{p-1}}{x^{p-\frac{p}{q}}} \\
&= \frac{p}{q}x^{p-1-p+\frac{p}{q}} \\
&= \frac{p}{q}x^{\frac{p}{q}-1} \\
&= rx^{r-1}.
\end{aligned}
$$

Negative Rational r. Let $f(x) = x^r$ where r is negative and rational, then there are positive integers, p and q so that $r = \frac{-p}{q}$, and thus $f(x) = x^{\frac{-p}{q}}$. Then

$$
\begin{aligned}
f'(x) &= \lim_{u\to x} \frac{f(u)-f(x)}{u-x} \\
&= \lim_{u\to x} \frac{u^r - x^r}{u-x} \\
&= \lim_{u\to x} \frac{u^{-p/q} - x^{-p/q}}{u-x} \\
&= \lim_{u\to x} \frac{\frac{1}{u^{p/q}} - \frac{1}{x^{p/q}}}{u-x} \\
&= \lim_{u\to x} \frac{\frac{x^{p/q}}{u^{p/q}x^{p/q}} - \frac{u^{p/q}}{u^{p/q}x^{p/q}}}{u-x} \\
&= \lim_{u\to x} \frac{x^{p/q}-u^{p/q}}{(u-x)\left(u^{p/q}x^{p/q}\right)} \\
&= \lim_{u\to x} \frac{-1\left(u^{p/q}-x^{p/q}\right)}{(u-x)\left(u^{p/q}x^{p/q}\right)} \\
&= \lim_{u\to x} \frac{-1}{\left(u^{p/q}x^{p/q}\right)}\frac{\left(u^{p/q}-x^{p/q}\right)}{(u-x)}.
\end{aligned}
$$

At this point, our limit is the same as in the positive rational case except for the factor $\frac{-1}{u^{p/q}x^{p/q}}$, and so making the same algebraic manipulations, we have

$$
\begin{aligned}
f'(x) &= \lim_{u\to x} \frac{-1}{\left(u^{p/q}x^{p/q}\right)}\frac{\left(u^{p/q}-x^{p/q}\right)}{(u-x)} \\
&= \lim_{u\to x} \frac{-1}{\left(u^{p/q}x^{p/q}\right)}\frac{\left(u^{p-1}+u^{p-2}x+u^{p-3}x^2+\cdots+x^{p-1}\right)}{\left(\left(u^{p/q}\right)^{q-1}+\left(u^{p/q}\right)^{q-2}\left(x^{p/q}\right)+\left(u^{p/q}\right)^{q-3}\left(x^{p/q}\right)^2+\cdots+\left(u^{p/q}\right)\left(x^{p/q}\right)^{q-2}+\left(x^{p/q}\right)^{q-1}\right)} \\
&= \frac{-1}{\left(x^{p/q}x^{p/q}\right)}\frac{px^{p-1}}{q\left(x^{p/q}\right)^{q-1}} \\
&= \frac{-p}{q}\frac{1}{x^{2p/q}}\frac{x^{p-1}}{x^{p-p/q}}
\end{aligned}
$$

$$= \frac{-p}{q} x^{p-1-p+p/q-2p/q}$$
$$= \frac{-p}{q} x^{-p/q-1}$$
$$= rx^{r-1}.$$

Irrational r. We now examine the case where r is not rational. We cannot calculate this derivative directly, but have to use the fact that the rational numbers are **dense** in the real numbers. Essentially, this means that there must be a sequence or rational numbers, $r_1, r_2, r_3, \ldots$ so that $r_k \xrightarrow{k\to\infty} r$.

We can then define function $f_k(x) = x^{r_k}$, and note that $x^{r_k} \xrightarrow{k\to\infty} x^r$, and by continuity,

$$\begin{aligned} f'(x) &= \lim_{k\to\infty} f_k'(x) \\ &= \lim_{k\to\infty} r_k x^{r_k-1} \\ &= rx^{r-1}. \end{aligned}$$

■

Theorem 4.2.4 Sums and Constant Multiples of Derivatives. *Let $f_1(x)$ and $f_2(x)$ be any functions that have derivatives on a shared domain, and let k_1 and k_2 be any (constant) real numbers. Then*

$$(k_1 f_1(x) + k_2 f_2(x))' = k_1 f_1'(x) + k_2 f_2'(x).$$

Checkpoint 4.2.5 Find the derivative of $f(x) = 4x^{30} - 5x^{17} + 15x^2 + 45$.

4.3 Derivative Shortcuts

4.3.1 Basic Functions

Theorem 4.3.1 Constant Functions. *For any constant, c,*

$$\text{if } f(x) = c, \qquad \text{then } f'(x) = 0$$

Theorem 4.3.2 Power Functions. *For any constant, $r \neq 0$,*

$$\text{if } f(x) = x^r, \qquad \text{then } f'(x) = rx^{r-1}$$

Theorem 4.3.3 Sine.

$$\text{If } f(x) = \sin(x), \qquad \text{then } f'(x) = \cos(x)$$

Theorem 4.3.4 Cosine.

$$\text{If } f(x) = \cos(x), \qquad \text{then } f'(x) = -\sin(x)$$

Theorem 4.3.5 Natural Exponential.

$$\text{If } f(x) = \mathrm{e}^x, \qquad \text{then } f'(x) = \mathrm{e}^x$$

4.3.2 Combinations of Functions

Theorem 4.3.6 Constant Multiple. *For any constant c and any differentiable function $f(x)$,*

$$\text{if } g(x) = c \cdot f(x), \qquad \text{then } g'(x) = c \cdot f'(x)$$

Theorem 4.3.7 Sums and Differences. *For any differentiable functions $f_1(x)$ and $f_2(x)$,*

$$\begin{aligned} &\text{if } g(x) = f_1(x) + f_2(x), && \text{then } g'(x) = f_1'(x) + f_2'(x) \\ &\text{if } g(x) = f_1(x) - f_2(x), && \text{then } g'(x) = f_1'(x) - f_2'(x) \end{aligned}$$

Theorem 4.3.8 Products. *For any differentiable functions $f_1(x)$ and $f_2(x)$,*

$$\text{if } g(x) = f_1(x) \cdot f_2(x), \qquad \text{then } g'(x) = f_1'(x) \cdot f_2(x) + f_1(x) \cdot f_2'(x)$$

Theorem 4.3.9 Quotients. *For any differentiable functions $f_1(x)$ and $f_2(x)$, for any x where $g(x) \neq 0$,*

$$\text{if } g(x) = \frac{f_1(x)}{f_2(x)} \qquad \text{then } g'(x) = \frac{f_1'(x) \cdot f_2(x) - f_1(x) \cdot f_2'(x)}{\big(f_2(x)\big)^2}$$

Theorem 4.3.10 Compositions. *For any differentiable functions $f_o(x)$ and $f_i(x)$,*

$$\text{if } g(x) = f_o\big(f_i(x)\big), \qquad \text{then } g'(x) = f_o'\big(f_i(x)\big) \cdot f_i'(x)$$

4.3.3 Justifications

4.3.3.1 Products

Let $f(x)$ and $g(x)$ be differentiable functions (with the same domain), and let $P(x) = f(x)g(x)$. Then

$$\begin{aligned} \big(f(x)g(x)\big)' &= P'(x) \\ &= \lim_{u \to x} \frac{P(u) - P(x)}{u - x} \\ &= \lim_{u \to x} \frac{f(u)g(u) - f(x)g(x)}{u - x} \end{aligned}$$

$$
\begin{aligned}
&= \lim_{u\to x} \frac{f(u)g(u) - f(x)g(u) + f(x)g(u) - f(x)g(x)}{u-x} \\
&= \lim_{u\to x} \frac{f(u)g(u) - f(x)g(u)}{u-x} + \frac{f(x)g(u) - f(x)g(x)}{u-x} \\
&= \lim_{u\to x} \left(\frac{f(u)-f(x)}{u-x}\right) g(u) + f(x)\left(\frac{g(u)-g(x)}{u-x}\right) \\
&= \lim_{u\to x} \left(\frac{f(u)-f(x)}{u-x}\right) g(u) + \lim_{u\to x} f(x)\left(\frac{g(u)-g(x)}{u-x}\right) \\
&= \lim_{u\to x} \left(\frac{f(u)-f(x)}{u-x}\right) \lim_{u\to x} g(u) + \lim_{u\to x} f(x) \lim_{u\to x}\left(\frac{g(u)-g(x)}{u-x}\right) \\
&= f'(x)g(x) + f(x)g'(x).
\end{aligned}
$$

4.3.3.2 Quotients

Let $f(x)$ and $g(x)$ be differentiable functions (with the same domain) with $g(x) \neq 0$ on the domain, and let $P(x) = \frac{f(x)}{g(x)}$. Then

$$
\begin{aligned}
\left(\frac{f(x)}{g(x)}\right)' &= P'(x) \\
&= \lim_{u\to x} \frac{P(u) - P(x)}{u-x} \\
&= \lim_{u\to x} \frac{\frac{f(u)}{g(u)} - \frac{f(x)}{g(x)}}{u-x} \\
&= \lim_{u\to x} \frac{1}{u-x}\left(\frac{f(u)}{g(u)} - \frac{f(x)}{g(x)}\right) \\
&= \lim_{u\to x} \frac{1}{u-x}\left(\frac{f(u)}{g(u)}\frac{g(x)}{g(x)} - \frac{f(x)}{g(x)}\frac{g(u)}{g(u)}\right) \\
&= \lim_{u\to x} \frac{1}{u-x}\left(\frac{f(u)g(x) - f(x)g(u)}{g(x)g(u)}\right) \\
&= \lim_{u\to x} \frac{1}{g(x)g(u)}\left(\frac{f(u)g(x) - f(x)g(u)}{u-x}\right) \\
&= \lim_{u\to x} \frac{1}{g(x)g(u)}\left(\frac{f(u)g(x) - f(x)g(x) + f(x)g(x) - f(x)g(u)}{u-x}\right) \\
&= \lim_{u\to x} \frac{1}{g(x)g(u)}\left(\left(\frac{f(u)g(x) - f(x)g(x)}{u-x}\right) + \left(\frac{f(x)g(x) - f(x)g(u)}{u-x}\right)\right) \\
&= \lim_{u\to x} \frac{1}{g(x)g(u)}\left(\left(\frac{f(u)-f(x)}{u-x}\right) g(x) + f(x)\left(\frac{g(x)-g(u)}{u-x}\right)\right) \\
&= \lim_{u\to x} \frac{1}{g(x)g(u)}\left(\left(\frac{f(u)-f(x)}{u-x}\right) g(x) - f(x)\left(\frac{g(u)-g(x)}{u-x}\right)\right) \\
&= \lim_{u\to x} \frac{1}{g(x)g(u)}\left(\lim_{u\to x}\left(\frac{f(u)-f(x)}{u-x}\right) g(x) - \lim_{u\to x} f(x)\left(\frac{g(u)-g(x)}{u-x}\right)\right) \\
&= \frac{1}{g(x)g(x)}\left(f'(x)g(x) - f(x)g'(x)\right) \\
&= \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}
\end{aligned}
$$

4.3.3.3 Compositions

Let $f(x)$ and $g(x)$ be differentiable functions, and let $C(x) = f\big(g(x)\big)$. Then, if we let $U = g(u)$ and $X = g(x)$, we have $U \xrightarrow{u\to x} X$, and so

$$\begin{aligned}
\big(f\big(g(x)\big)\big)' &= C'(x) \\
&= \lim_{u\to x} \frac{C(u) - C(x)}{u - x} \\
&= \lim_{u\to x} \frac{f\big(g(u)\big) - f\big(g(x)\big)}{u - x} \\
&= \lim_{u\to x} \frac{f\big(g(u)\big) - f\big(g(x)\big)}{g(u) - g(x)} \cdot \frac{g(u) - g(x)}{u - x} \\
&= \lim_{g(u)\to g(x)} \frac{f\big(g(u)\big) - f\big(g(x)\big)}{g(u) - g(x)} \cdot \lim_{u\to x} \frac{g(u) - g(x)}{u - x} \\
&= \lim_{U\to X} \frac{f\big(U\big) - f\big(X\big)}{U - X} \cdot \lim_{u\to x} \frac{g(u) - g(x)}{u - x} \\
&= \lim_{U\to X} \frac{f\big(U\big) - f\big(X\big)}{U - X} \cdot \lim_{u\to x} \frac{g(u) - g(x)}{u - x} \\
&= f'\left(X\right) \cdot g'(x) \\
&= f'\big(g(x)\big) \cdot g'(x)
\end{aligned}$$

4.3.4 Consequences

Example 4.3.11 The Derivative of $\tan(x)$**.** Let consider $f(x) = \tan(x) = \frac{\sin(x)}{\cos(x)}$. Then

$$\begin{aligned}
f'(x) &= \frac{(\sin(x))'(\cos(x)) - (\sin(x))(\cos(x))'}{\cos^2(x)} \\
&= \frac{(\cos(x))(\cos(x)) - (\sin(x))(-\sin(x))}{\cos^2(x)} \\
&= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} \\
&= \frac{1}{\cos^2(x)} \\
&= \sec^2(x)
\end{aligned}$$

or, equivalently

$$\begin{aligned}
f'(x) &= \frac{\cos^2(x)}{\cos^2(x)} + \frac{\sin^2(x)}{\cos^2(x)} \\
&= 1 + \tan^2(x).
\end{aligned}$$

▲

Example 4.3.12 The Derivative of $\sec(x)$**.** Let $f(x) = \sec(x) = \frac{1}{\cos(x)}$. Then

$$\begin{aligned}
f'(x) &= \frac{(1)'(\cos(x)) - (1)(\cos(x))'}{\cos^2(x)} \\
&= \frac{(0)(\cos(x)) - (1)(-\sin(x))}{\cos^2(x)} \\
&= \frac{\sin(x)}{\cos^2(x)}
\end{aligned}$$

$$= \frac{1}{\cos(x)} \frac{\sin(x)}{\cos(x)}$$
$$= \sec(x) \tan(x).$$

▲

4.4 Skills Check 0

1. $f(x) = 7x^8 - 3x^2 - 4x - 13$

2. $g(x) = \big(\tan(x)\big)^{1/5}$

3. $q(u) = \big(\sin(4u^2 - 1)\big)^3$

4. $h(x) = \dfrac{(x^2 - 3x)^2}{x^3 - 4}$

5. $p(r) = \dfrac{2r + 3}{2r^3 + 1}$

6. $m(t) = e^{3\cos(2t-1)}$

7. $r(x) = (-2\sqrt[4]{x} + 4x^5)(x^2 - 3x^3 + 1)$

8. $T(z) = \sin^2(3z + 1)$

9. $g(w) = 3^{\sqrt{4w^2-1}}$

10. $F(x) = \dfrac{1}{2x^3} - 3x^{-1/3}$

4.5 Skills Check 1

1. $f(x) = 8x^5 - 12x^3 + 5x^2 + 7$

2. $g(x) = \sqrt[5]{\cos(x)}$

3. $q(t) = \cos^2(4t^3 + 3t^2)$

4. $p(r) = \dfrac{r^2 + 2r + 3}{3r + 2}$

5. $h(x) = \dfrac{x^3 - x^2 + 1}{(x^2 + 2)^4}$

6. $m(t) = \mathrm{e}^{\sqrt{2t+3}}$

7. $r(x) = (\sqrt[3]{x} - x^3)\left(\dfrac{1}{3x} + \dfrac{x}{3}\right)$

8. $T(z) = z^5 \tan(z)$

9. $g(w) = \mathrm{e}^{\sin(3w^2)}$

10. $F(x) = \dfrac{1}{3x^4} - 5x^{-1/2}$

Chapter 5

Progress Checks

5.1 Progress Check 1

1. An object is moving in such a way that its position s (in meters) at time t (in seconds) is given by the function $s(t) = 8t^2 - 10t + 100$.

(a) **(i)** Set up, *but do not evaluate*, the limit definition of the derivative that would find the derivative of the function at $t = 1$.

(ii) Clearly explain why the limit you set up in Part a, p. 80 will find the derivative of $s(t)$ at $t = 1$.

(b) Evaluate the limit you found in Part a, p. 80.

(c) Find an equation for the tangent line to the graph of $s(t)$ from Question 1, p. 80 at the point $t = 1$.

2. A company's monthly revenue (money taken in) from car sales, R (in thousands of dollars), is a function of monthly advertising spending, a (in thousands of dollars). That is, for some function f, $R = f(a)$.

(a) Interpret the following in practical terms, paying close attention to units.

$$f(80) = 250.$$

(b) What does the company expect is true about the sign of f'? Why?

(c) What does the statement $f'(100) = 2$ mean in practical terms?

(d) What does the statement $f'(100) = 0.5$ mean in practical terms?

(e) Suppose the company currently spends about \$100,000 per month on advertising.

(i) If $f'(100) = 2$, should the company spend more or less than \$100,000 on advertising?

(ii) If $f'(100) = 0.5$, should the company spend more or less than \$100,000 on advertising?

3. A water tank in the shape of a cone with the point upward with a base radius of 6 ft and a height of 24 ft, as shown in Figure 5.1.1, p. 82, is filling at a constant rate of 0.12 ft^3/min. The goal of this problem is to construct a model for the depth of water as a function of time.

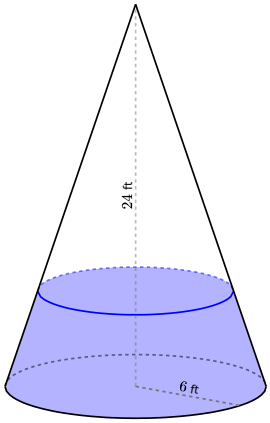


Figure 5.1.1 A Conical Tank with Base Radius 6 ft and Height 24 ft.

(a) Find the total volume of the tank in cubic feet.

(b) Find the volume of water in the tank (in cubic feet) when the tank is filled to a depth of 2 ft and when the tank is filled to a depth of 4 ft.

(c) Find a formula for the volume of water in the tank (in cubic feet) when the tank is filled to a depth of d ft.

Hint. The volume of water is the volume of the whole tank minus the volume of the empty cone (from the surface of the water to the top of the tank).

(d) On the blank grid in Figure 5.1.2, p. 83, sketch a graph of the depth of water *as a function of time*. Your sketch does not need to be perfect, but it should reflect how the rate of change of depth is changing as the tank fills.

You may assume the tank is empty to begin.

Figure 5.1.2 A Blank Grid for Sketching the Graph of Depth as a Function of Time

(e) Find an expression for the volume of water in the tank as a function of time, t, in minutes. (You may assume the tank is empty to begin.)

(f) Equate the volume in terms of time with the volume in terms of depth and solve for depth. This yields a formula for the depth as a function of time.

4. A water tank in the shape of a horizontal cylinder with a radius of 3 m and a length of 8 m, as shown in Figure 5.1.3, p. 84, is filling at a constant rate of $1.1\text{m}^3/\text{min}$. The goal of this problem is to approximate instantaneous rate of change numerically when the water is 2m deep.

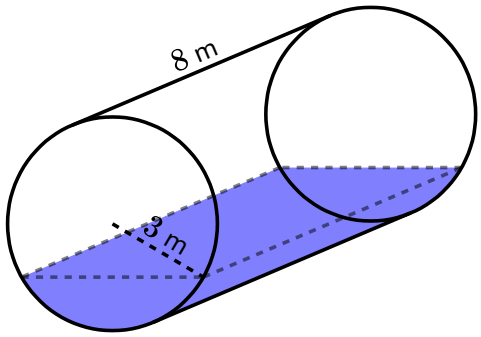


Figure 5.1.3 A Cylindrical Tank with Radius 3 m and Length 8 m.

(a) Find the total volume of the tank in cubic meters.

(b) Find the volume of water in the tank (in cubic meters) when the tank is filled to a depth of 3 m.

(c) How long will it take for the tank to fill to a depth of 3 m if it starts empty?

(d) How long will it take for the tank to fill completely if it starts empty?

(e) On the blank grid in Figure 5.1.4, p. 85, sketch a graph of the depth of water *as a function of time*. Your sketch does not need to be perfect, but it should reflect how the rate of change of depth is changing as the tank fills.

You may assume the tank is empty to begin.

Figure 5.1.4 A Blank Grid for Sketching the Graph of Depth as a Function of Time

(f) The volume of water in the tank, in cubic meters, as a function of depth, h, in meters, is given by

$$V(h) = 8\,(h-3)\,\sqrt{9-(h-3)^2} + 72\arcsin\left(\frac{h-3}{3}\right) + 36\pi.$$

(i) Use the volume function, $V(h)$, to find the volume of water in the tank at a depth of 1.9 m and at a depth of 2.1 m.

(ii) Use the fill rate to find how long it would take for the tank to fill from a depth of 1.9 m to a depth of 2.1 m.

(iii) Calculate the average rate of change of depth as a function of time between the times when the depth is 1.9 m when the depth is 2.1 m.

Be careful! Your result should have units meters per minute!

(g) Repeat the previous problem with depths of 1.99 and 2.01 meters.

5.2 Progress Check 2

1. Interpreting Derivatives and Linear Approximation. In this problem, you will investigate how speed (in miles per hour) and the total time for a ten mile trip are related.

The total time (in seconds) for a trip of 10 miles at a constant speed of $v$mph (where $v > 0$) is given by

$$\begin{aligned} T(v) &= \frac{10\text{mi}}{v\text{mph}} \cdot \frac{3600\text{s}}{1\text{hr}} \\ &= \frac{36,000}{v} \\ &= 36,000v^{-1}. \end{aligned}$$

(a) Calculate and interpret (including units) $T(60)$.

(b) Based on the physical context, would you expect $T'(v)$ to be positive or negative? Explain.

(c) Calculate $T'(v)$. (You may use either the limit definition of derivative or a shortcut that we have learned.)

(d) Calculate and interpret (including units) $T'(60)$.

(e) **(i)** Find an equation for the linear approximation of T at $v = 60$. Call this function $L_T(v)$.

(ii) Use your linear approximation to approximate $T(65)$.

(f) **(i)** Calculate and interpret (including units) $T(65)$.

(ii) How does the actual value of $T(65)$ compare with the value from the linear approximation? What is the magnitude of the error, and is the linear approximation an over- or under-estimate?

(g) Compare the value of the linear approximation at 61, $L_T(61)$, with the value of $T(61)$. What is the magnitude of the error, and is the linear approximation an over- or under-estimate? How does this compare to the error at $v = 65$?

(h) Calculate and interpret (including units) $T'(75)$.

(i) Would you expect $T''(v)$ to be positive or negative? Explain.

2. Modeling and Optimization. A trough is made from a 5ft × 20ft sheet of metal by folding two 1ft × 20ft sections up at an angle θ and attaching trapezoidal caps at the ends. A cross-section is shown in Figure 1, p. 88.

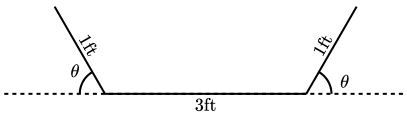


Figure 5.2.1 Cross-Section of a Trough

(a) Express the total cross-sectional area, $A(\theta)$, and then the total volume of the trough, $V(\theta)$, in terms of trigonometric functions of θ.

Hint. Draw a vertical line through each of the corners on the base to divide the cross-section into a rectangle and two right triangles.

The total volume will be cross-sectional area times 20ft.

(b) Calculate $V'(\theta)$.

(c) Find the angle θ that maximizes the volume of the trough.

Hint. What will be true about the derivative at the point where the volume is highest?

You will need to use the identity $\cos^2(\theta) + \sin^2(\theta) = 1$ and solve a quadratic equation for $\cos(\theta)$.

3. Figure 2, p. 89 shows the graphs $y = f(x)$ (solid) and $y = g(x)$ (dashed), one of which is the derivative of the other.

Determine which graph is the derivative and justify your answer in terms of the relationships between a function and its derivative.

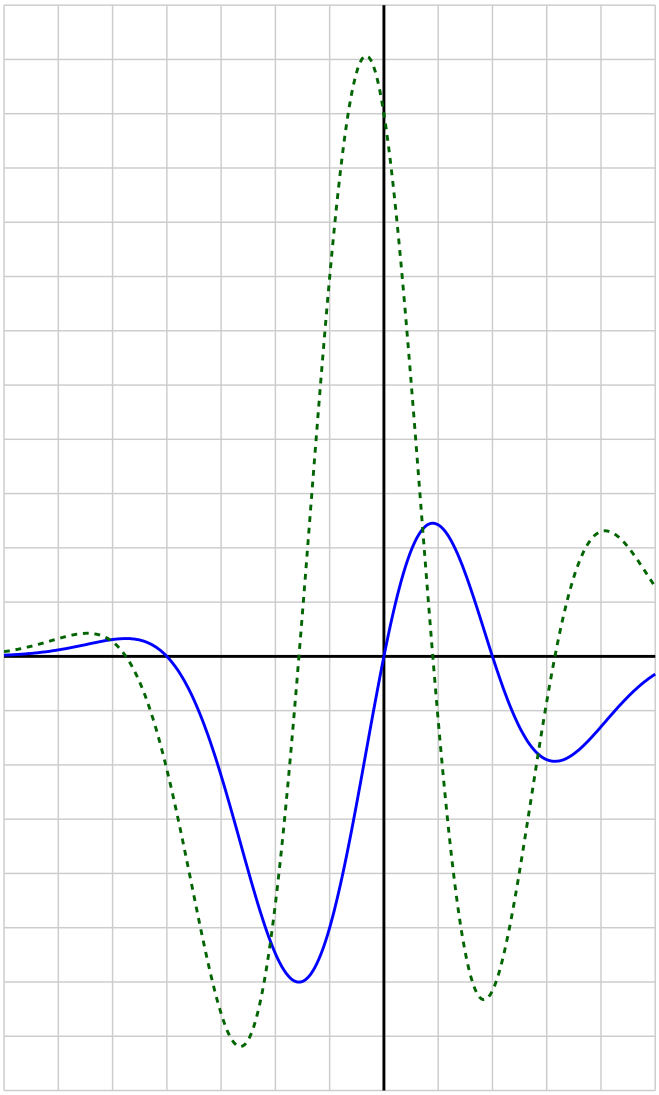


Figure 5.2.2 The graphs of $f(x)$ and $g(x)$.

4. Computation and Patterns. In this problem, you will calculate several derivatives (using shortcuts) and look for patterns in the derivatives.

(a) *Squares of Functions.*

(i) Calculate the derivative of x^6 in two ways: first by using the pattern for derivatives of x^r directly, and then using the fact that $x^6 = \left(x^3\right)\left(x^3\right)$.

(ii) Calculate the derivative of $\mathrm{e}^{2x} = \left(\mathrm{e}^x\right)^2$ using the fact that $\mathrm{e}^{2x} = \mathrm{e}^x \cdot \mathrm{e}^x$.

(iii) Calculate the derivative of $\cos^2(x)$ using the fact that $\cos^2(x) = \cos(x) \cdot \cos(x)$.

(iv) Let $f(x)$ be a differentiable function (with derivative $f'(x)$) and calculate the derivative of $\left(f(x)\right)^2 = f(x) \cdot f(x)$. Compare this result to your previous calculations.

(b) *Cubes of Functions.*

(i) Calculate the derivative of $\mathrm{e}^{3x} = \left(\mathrm{e}^x\right)^3$ using your previous work and the fact that $\mathrm{e}^{3x} = \mathrm{e}^{2x} \cdot \mathrm{e}^x$.

(ii) Calculate the derivative of $\cos^3(x)$ using your previous work and the fact that $\cos^3(x) = \cos^2(x) \cdot \cos(x)$.

(iii) Let $f(x)$ be a differentiable function (with derivative $f'(x)$) and calculate the derivative of $\left(f(x)\right)^3 = \left(f(x)\right)^2 \cdot f(x)$.
Compare this result to your previous calculations.